Supplementary Material 2: Additional analyses using a correction in the number of births based on under-registration estimates

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Contents

1 Introduction

This document aims to redo the analyses made for the article "Fertility rate among adolescents in Brazil in the period 2012-2021: effects of human development, primary care coverage and the COVID-19 pandemic" using a correction for the numbers of live births based on under-registration estimates, aiming to show that the original analyses are not significantly affected by the lack of this correction.

2 Importing the necessary packages

```r
library(knitr)
library(kableExtra)
library(reactable)
library(dplyr)
library(glue)
library(tidyr)
library(ggplot2)
library(trend)
library(factoextra)
library(clusterSim)
library(clValid)
library(htmltools)
library(gridExtra)
library(grid)
library(rstatix)
library(sf)
library(geobr)
library(corrplot)
library(gamlss)
```

3 About the dataset

```r
# Importing the dataset
df_indicadores <- read.csv("databases/data_fertility_rates_article.csv") |>
  mutate(
    regiao = factor(
      regiao,
      levels = c("Norte", "Nordeste", "Centro-Oeste", "Sudeste", "Sul")
    )
  )
```

3.1 Variables description

The dataset used in this study included variables available on several different databases. To obtain the variables from SINASC (more specifically, the number of births of mothers in each age group), we used the `{microdatasus}` package in R. To obtain the population estimates for each age group, we used Tabnet DATASUS (using the updated estimates after the 2022 census). To obtain the number of women beneficiaries of private health plans (used for calculating the percentage of women who exclusively use the SUS), we used Tabnet ANS. To obtain the variables used to calculate the Primary Care coverage, we used the "e-Gestor Atenção Primária à Saúde" platform. Finally, to obtain the HDI-M, we used Atlas Brasil. All data was obtained for the period 2012 to 2021. A description of the variables included in the file `data_fertility_rates_article.csv` can be seen in **Table 1**.

Table 1: Description of the variables in the dataset.

Variable	Description	Source
codmunres	Municipality code	SINASC
municipio	Municipality name	SINASC
uf	Federative Unit	SINASC
regiao	Brazilian Region	SINASC
ano	Year	—
nvm_10_a_14	Live births of mothers aged 10 to 14 years	SINASC
nvm_15_a_19	Live births of mothers aged 15 to 19 years	SINASC
nvm_10_a_19	Live births of mothers aged 10 to 19 years	SINASC
pop_feminina_10_a_14	Estimated female population aged 10 to 14 years	Tabnet DATASUS
pop_feminina_15_a_19	Estimated female population aged 15 to 19 years	Tabnet DATASUS
pop_feminina_10_a_19	Estimated female population aged 10 to 19 years	Tabnet DATASUS
pop_feminina_10_a_49	Estimated female population aged 10 to 49 years	Tabnet DATASUS
idhm	HDI-M	Atlas Brasil
media_cobertura_ab	Population covered by Primary Care teams	e-Gestor Atenção Primária à Saúde
populacao_total	Total population	e-Gestor Atenção Primária à Saúde
pop_fem_10_49_com_plano_saude	Female population aged 10 to 49 years with private health insurance	Tabnet ANS

3.2 Missing data

The only variable in the data set that contained missing information was the HDI-M. The five municipalities with missing HDI-M were removed from the dataset only for the modeling phase.

```r
# Checking for NAs
colSums(is.na(df_indicadores))
```

```
##                    codmunres                    municipio
##                            0                            0
##                           uf                       regiao
##                            0                            0
##                          ano                  nvm_10_a_14
##                            0                            0
##                  nvm_15_a_19                  nvm_10_a_19
##                            0                            0
##          pop_feminina_10_a_14         pop_feminina_15_a_19
##                            0                            0
##          pop_feminina_10_a_19         pop_feminina_10_a_49
##                            0                            0
##                         idhm          media_cobertura_ab
##                           50                            0
##              populacao_total pop_fem_10_49_com_plano_saude
##                            0                            0
```

```r
# Checking which municipalities had missing HDI-M
unique(df_indicadores$municipio[is.na(df_indicadores$idhm)])
```

```
## [1] "Mojuí dos Campos"  "Pescaria Brava"    "Balneário Rincão"
## [4] "Pinto Bandeira"    "Paraíso das Águas"
```

3.3 Municipalities with zero births and under-registration

As it can be seen in **Table 2**, all the years in the considered period included municipalities that had no births from mothers in each age group.

```r
# Checking how many municipalities had 0 births from mothers in each age group per year
kbl(
  df_indicadores |>
  group_by(ano) |>
  summarise(
    municipalities_zero_births_10_to_14 = sum(nvm_10_a_14 == 0, na.rm = TRUE),
    municipalities_zero_births_15_to_19 = sum(nvm_15_a_19 == 0, na.rm = TRUE),
    municipalities_zero_births_10_to_19 = sum(nvm_10_a_19 == 0, na.rm = TRUE)
  ) |>
  ungroup(),
  col.names = c(
    "Year",
    "Municipalities with zero births (10-14)",
    "Municipalities with zero births (15-19)",
    "Municipalities with zero births (10-19)"
  ),
  align = "cccc",
  caption = "Municipalities with zero births in each year for each age group (before
    the correction)."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(1:4, width = "4cm")
```

Table 2: Municipalities with zero births in each year for each age group (before the correction).

Year	Municipalities with zero births (10-14)	Municipalities with zero births (15-19)	Municipalities with zero births (10-19)
2012	1704	37	36
2013	1754	42	40
2014	1735	40	39
2015	1781	34	34
2016	1860	46	43
2017	1945	57	52
2018	2010	45	39
2019	2116	59	58
2020	2212	66	59
2021	2196	80	76

In addition, as it can be seen in **Table 3**, 227 municipalities had no births from mothers aged 10 to 14 in the entire period.

```
# Checking how many municipalities had 0 births in the whole period for each age group
kbl(
  df_indicadores |>
  group_by(codmunres) |>
  summarise(
    all_zeros_10_to_14 = all(nvm_10_a_14 == 0, na.rm = TRUE),
    all_zeros_15_to_19 = all(nvm_15_a_19 == 0, na.rm = TRUE),
    all_zeros_10_to_19 = all(nvm_10_a_19 == 0, na.rm = TRUE)
  ) |>
  summarise(
    municipalities_with_all_zeros_10_to_14 = sum(all_zeros_10_to_14),
    municipalities_with_all_zeros_15_to_19 = sum(all_zeros_15_to_19),
    municipalities_with_all_zeros_10_to_19 = sum(all_zeros_10_to_19)
  ) |>
  ungroup(),
  col.names = c(
    "Municipalities with zero births in the while period (10-14)",
    "Municipalities with zero births in the while period (15-19)",
    "Municipalities with zero births in the while period (10-19)"
  ),
  align = "ccc",
  caption = "Municipalities with zero births in the entire period for each age group
    (before the correction)."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(1:3, width = "4cm")
```

Table 3: Municipalities with zero births in the entire period for each age group (before the correction).

Municipalities with zero births in the while period (10-14)	Municipalities with zero births in the while period (15-19)	Municipalities with zero births in the while period (10-19)
227	0	0

In an attempt to take into account factors such as under-registration of the number of live births in small municipalities, we tried to correct the numbers of births using the SINASC under-registration estimates provided by the IBGE. The output below summarizes the SINASC coverage statistics presented by IBGE for Brazilian municipalities from 2015 to 2021.

```
# Reading a dataset with SINASC under-registration estimates
df_cobertura_sinasc <- read.csv("databases/dados_cobertura_sinasc_2015_2021.csv")

df_cobertura_sinasc |>
  select(cobertura_sinasc) |>
  summary()
```

```
##  cobertura_sinasc
##  Min.   : 24.00
##  1st Qu.: 98.40
##  Median : 99.70
##  Mean   : 98.68
##  3rd Qu.:100.00
##  Max.   :100.00
```

The lowest coverage was observed in Condor (Rio Grande do Sul), with 24% in 2015. However, as we can see, the vast majority of municipalities exhibited very high coverage, with the first quartile at 98.4%. Futhermore, despite a small decrease in the number of municipalities that did not report births in each age group, as we can see in **Table 4.**, we decided not to use this approach for the main analyses because:

1) SINASC under-registration estimates are only available from 2015 onwards, preventing their use in the temporal analyses covering the entire period from 2012 to 2021;
2) not all municipalities contained under-registration estimates;
3) even for the analyses in which we could make this correction for the entire period considered (i.e the cluster analyses and modeling), new fitting attempts revealed that the correction had minimal influence on the results - for the modeling, for instance, the differences in the estimated coefficients were at most 5%, and the differences in the interpretations obtained were at most 2%.

For these reasons, and considering that the macro-level patterns remained unchanged, we opted to calculate the fertility rates in their raw form, without applying additional correction techniques. Therefore, this supplementary material aims to redo the original analyses, correcting the numbers of births using under-registration estimates, aiming to show that the results obtained are not significantly affected by this correction.

Note: since SINASC under-registration estimates are only available from 2015 onwards, the first stage of the article's analyses, in which we performed a temporal analysis of the period from 2012 to 2021, could not be reproduced in this supplementary material.

```
# Adjusting the number of births according to the under-registration estimates
df_indicadores_corrigido <- df_indicadores |>
  left_join(df_cobertura_sinasc) |>
  filter(ano >= 2015) |>
  mutate(
    nvm_10_a_14_corrigido = round(nvm_10_a_14 * 100 / cobertura_sinasc),
    nvm_15_a_19_corrigido = round(nvm_15_a_19 * 100 / cobertura_sinasc),
    nvm_10_a_19_corrigido = round(nvm_10_a_19 * 100 / cobertura_sinasc)
  ) |>
  mutate(
    nvm_10_a_14_corrigido = ifelse(is.na(nvm_10_a_14_corrigido), nvm_10_a_14, nvm_10_a_14_corrigido),
    nvm_15_a_19_corrigido = ifelse(is.na(nvm_15_a_19_corrigido), nvm_15_a_19, nvm_15_a_19_corrigido),
    nvm_10_a_19_corrigido = ifelse(is.na(nvm_10_a_19_corrigido), nvm_10_a_19, nvm_10_a_19_corrigido),
  )
```

```r
# Checking how many municipalities had 0 births in each age group per year after the correction
kbl(
  df_indicadores_corrigido |>
  group_by(ano) |>
  summarise(
    municipalities_zero_births_10_to_14 = sum(nvm_10_a_14_corrigido == 0, na.rm = TRUE),
    municipalities_zero_births_15_to_19 = sum(nvm_15_a_19_corrigido == 0, na.rm = TRUE),
    municipalities_zero_births_10_to_19 = sum(nvm_10_a_19_corrigido == 0, na.rm = TRUE)
  ) |>
  ungroup(),
  col.names = c(
    "Year",
    "Municipalities with zero births (10-14)",
    "Municipalities with zero births (15-19)",
    "Municipalities with zero births (10-19)"
  ),
  align = "cccc",
  caption = "Municipalities with zero births in each year for each age group (after
    the correction)."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(1:4, width = "4cm")
```

Table 4: Municipalities with zero births in each year for each age group (after the correction).

Year	**Municipalities with zero births (10-14)**	**Municipalities with zero births (15-19)**	**Municipalities with zero births (10-19)**
2015	1781	34	34
2016	1860	46	43
2017	1945	57	52
2018	2010	45	39
2019	2116	59	58
2020	2212	66	59
2021	2196	80	76

4 Phase 2: Cluster analysis

This stage of the analysis aimed to find four different groupings for Brazilian municipalities: one that took into account the fertility rate of women aged 10 to 14 in the period from 2018 to 2019; one that took into account this same rate, but in the period from 2020 to 2021; one that took into account the fertility rate of women aged 15 to 19 in the period from 2018 to 2019; and one that considered this same rate, but in the period from 2020 to 2021. To this end, we adjusted non-hierarchical methods (K-means and K-medoids) and hierarchical methods (Ward, single linkage, complete linkage and average linkage). For the non-hierarchical methods, we chose candidates for the number of groups using the knee plot. For the hierarchical methods, we chose candidates for the number of groups using dendrograms. We compared the chosen candidates using the Davies-Bouldin, Dunn, Silhouette and Calinski-Harabasz indices, and chose as final clusters those that presented the best average performance in these metrics (as long as the number of observations in each cluster is adequate). Finally, after determining the four different groupings, we compared, for each grouping, the behavior of some variables of interest in each cluster formed.

4.1 Preparing the data

```
# Preparing the data
## Filtering and calculating the fertility rates for each period
df_indicadores_18_19 <- df_indicadores_corrigido |>
  filter(ano %in% c(2018, 2019)) |>
  group_by(codmunres) |>
  summarise(
    tx_fecundidade_10_a_14 = round(
      sum(nvm_10_a_14_corrigido) / sum(pop_feminina_10_a_14) * 1000,
      1
    ),
    tx_fecundidade_15_a_19 = round(
      sum(nvm_15_a_19_corrigido) / sum(pop_feminina_15_a_19) * 1000,
      1
    ),
    cobertura_ab = round(
      sum(media_cobertura_ab) / sum(populacao_total) * 100,
      1
    ),
    porc_exclusivas_sus = round(
      sum(pop_fem_10_49_com_plano_saude) / sum(pop_feminina_10_a_49) * 100,
      1
    ),
    idhm = mean(idhm)
  ) |>
  ungroup()

df_indicadores_20_21 <- df_indicadores_corrigido |>
  filter(ano %in% c(2020, 2021)) |>
  group_by(codmunres) |>
  summarise(
    tx_fecundidade_10_a_14 = round(
      sum(nvm_10_a_14_corrigido) / sum(pop_feminina_10_a_14) * 1000,
      1
    ),
    tx_fecundidade_15_a_19 = round(
      sum(nvm_15_a_19_corrigido) / sum(pop_feminina_15_a_19) * 1000,
      1
    ),
    cobertura_ab = round(
      sum(media_cobertura_ab) / sum(populacao_total) * 100,
      1
    ),
    porc_exclusivas_sus = round(
      sum(pop_fem_10_49_com_plano_saude) / sum(pop_feminina_10_a_49) * 100,
      1
    ),
    idhm = mean(idhm)
  ) |>
  ungroup()

## Standardizing the data for each of the four cluster analysis
dados1_18_19 <- df_indicadores_18_19 |>
  select(tx_fecundidade_10_a_14) |>
  scale()
dados2_18_19 <- df_indicadores_18_19 |>
  select(tx_fecundidade_15_a_19) |>
```

```
  scale()

dados1_20_21 <- df_indicadores_20_21 |>
  select(tx_fecundidade_10_a_14) |>
  scale()
dados2_20_21 <- df_indicadores_20_21 |>
  select(tx_fecundidade_15_a_19) |>
  scale()

## Calculating the distance matrices
dados1_18_19_dist <- dist(dados1_18_19, method = "euclidean")
dados2_18_19_dist <- dist(dados2_18_19, method = "euclidean")

dados1_20_21_dist <- dist(dados1_20_21, method = "euclidean")
dados2_20_21_dist <- dist(dados2_20_21, method = "euclidean")
```

4.2 For the 10 to 14 age group

4.2.1 K-Means

4.2.1.1 2018-2019

The K-Means knee plot for the 2018-2019 period and 10-14 age group can be seen in **Figure 1**.

```
# Ploting the k-means knee-plot for the 2018-2019 period
fviz_nbclust(dados1_18_19, kmeans, method = "wss")
```

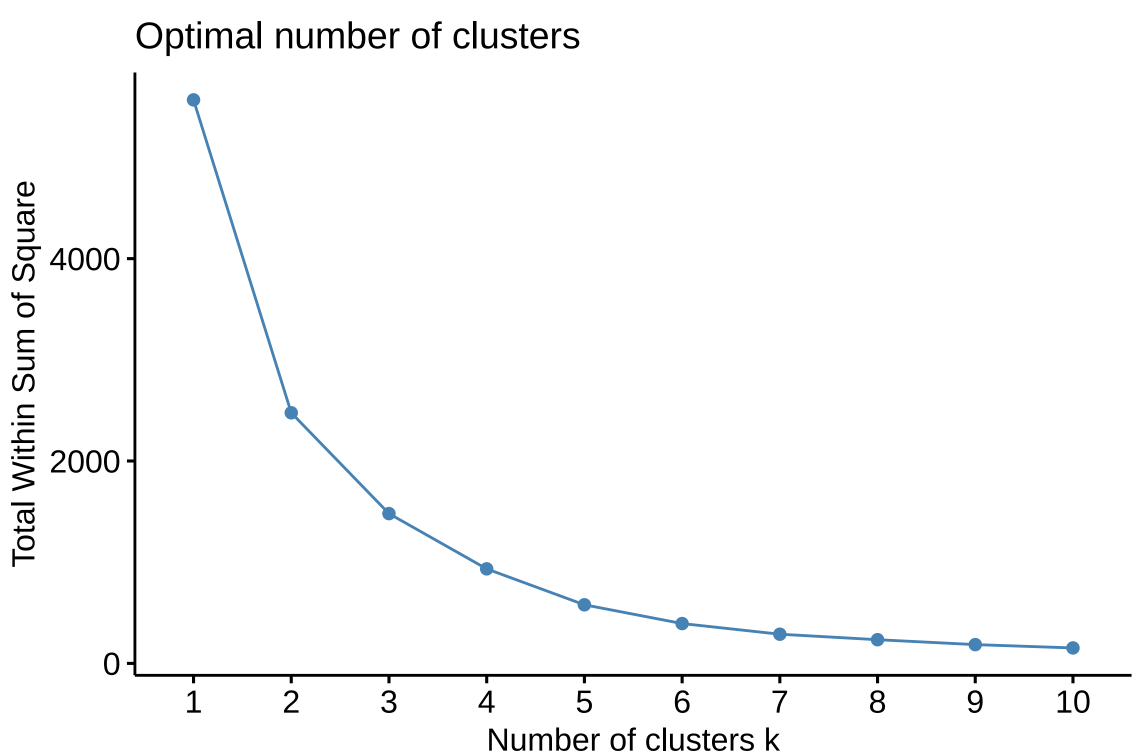

Figure 1: K-Means knee-plot for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d1_kmeans3_18_19 <- kmeans(dados1_18_19, 3)

set.seed(1504)
d1_kmeans4_18_19 <- kmeans(dados1_18_19, 4)
```

4.2.1.2 2020-2021

The K-Means knee plot for the 2020-2021 period and 10-14 age group can be seen in **Figure 2**

```
# Ploting the k-means knee-plot for the 2020-2021 period
fviz_nbclust(dados1_20_21, kmeans, method = "wss")
```

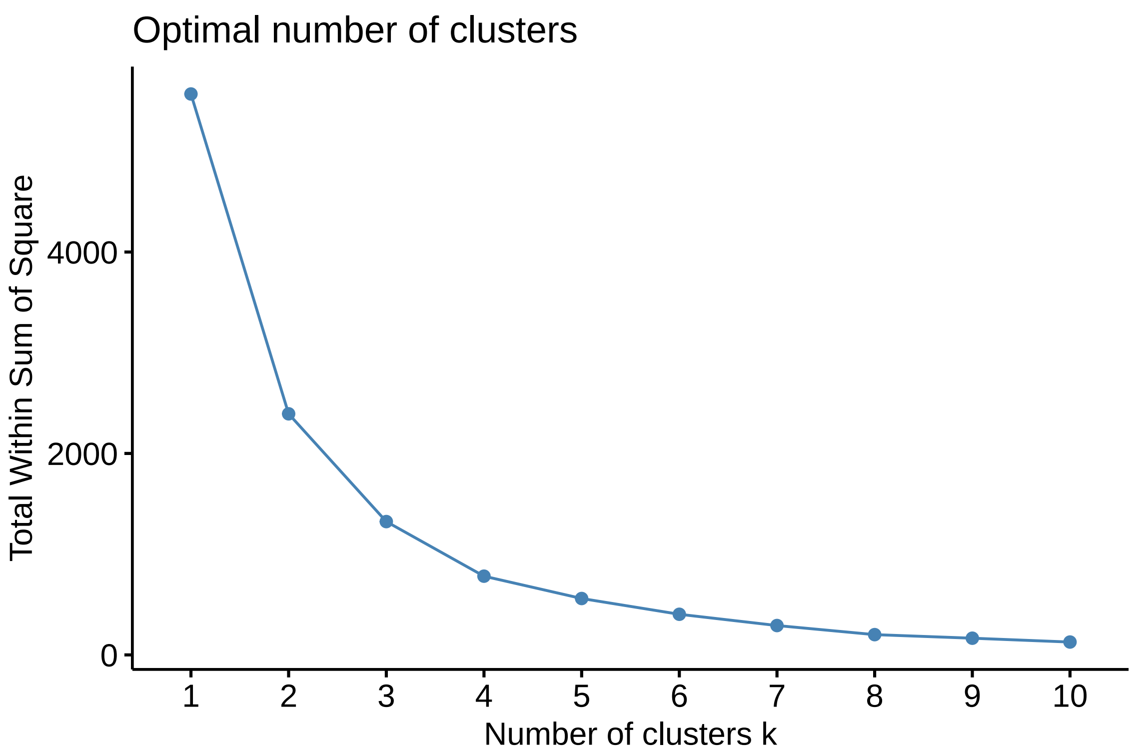

Figure 2: K-Means knee-plot for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d1_kmeans3_20_21 <- kmeans(dados1_20_21, 3)

set.seed(1504)
d1_kmeans4_20_21 <- kmeans(dados1_20_21, 4)
```

4.2.2 K-Medians

4.2.2.1 2018-2019

The K-Medians knee plot for the 2018-2019 period and 10 to 14 age group can be seen in **Figure 3**.

```
# Ploting the k-medians knee-plots for the 2018-2019 period
fviz_nbclust(dados1_18_19, cluster::pam, method = "wss")
```

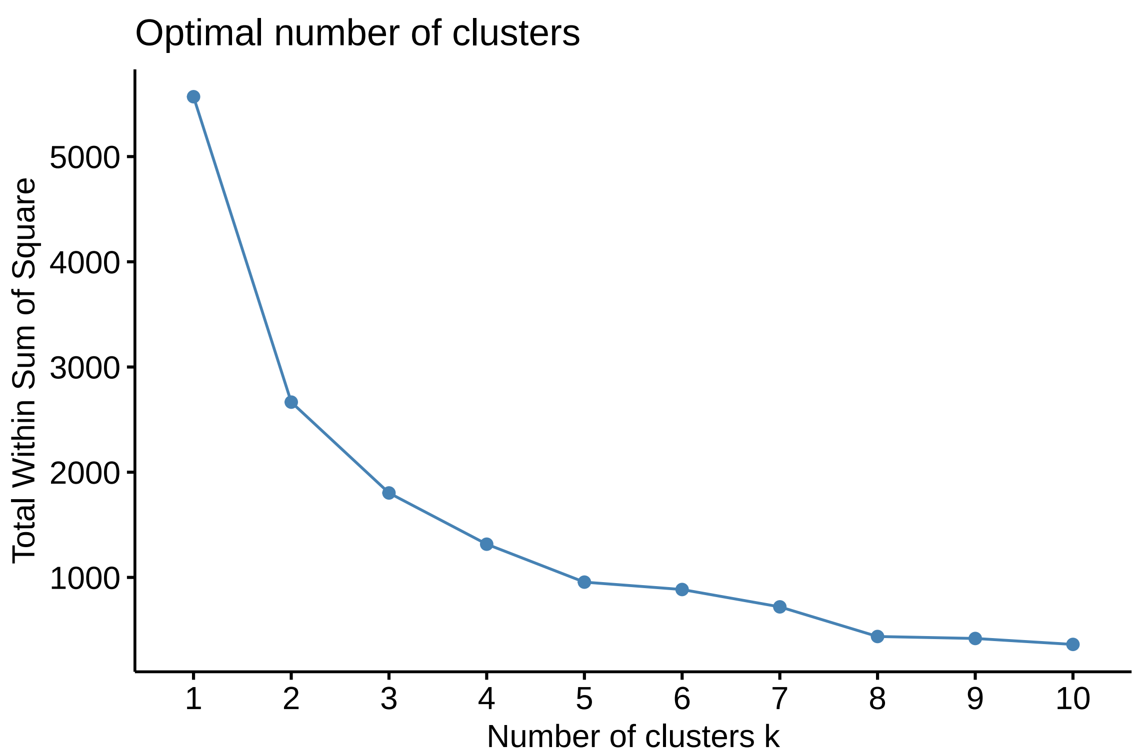

Figure 3: K-Medians knee-plot for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d1_kmedoids3_18_19 <- cluster::pam(dados1_18_19, k = 3)

set.seed(1504)
d1_kmedoids4_18_19 <- cluster::pam(dados1_18_19, k = 4)
```

4.2.2.2 2020-2021

The K-Medians knee plot for the 2020-2021 period and 10 to 14 age group can be seen in **Figure 4**.

```
# Ploting the k-medians knee-plots for the 2020-2021 period
fviz_nbclust(dados1_20_21, cluster::pam, method = "wss")
```

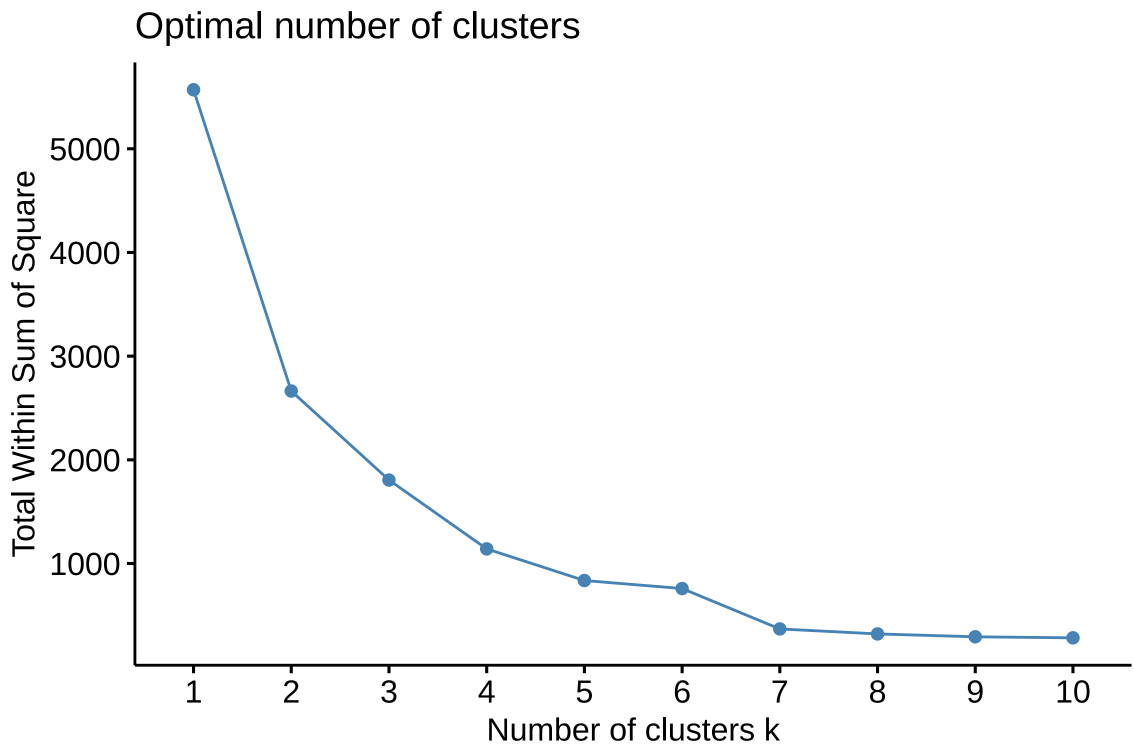

Figure 4: K-Medians knee-plot for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d1_kmedoids3_20_21 <- cluster::pam(dados1_20_21, k = 3)

set.seed(1504)
d1_kmedoids4_20_21 <- cluster::pam(dados1_20_21, k = 4)
```

4.2.3 Ward's method

4.2.3.1 2018-2019

The dendrogram for Ward's method for the 2018-2019 period and 10 to 14 age group can be seen in **Figure 5**.

```
# Ploting the drodrogram for the 2018-2019 period
set.seed(1504)
d1_ward_18_19 <- hclust(dados1_18_19_dist, method = "ward.D2")
plot(as.dendrogram(d1_ward_18_19))
```

Figure 5: Dendrogram for Ward's method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 3.

```
# Adjusting Ward's method with the chosen numbers of clusters
d1_ward3_18_19_class <- cutree(d1_ward_18_19, k = 3)
```

4.2.3.2 2020-2021

The dendrogram for Ward's method for the 2020-2021 period and 10 to 14 age group can be seen in **Figure 6**.

```
# Ploting the drodrogram for the 2020-2021 period
set.seed(1504)
d1_ward_20_21 <- hclust(dados1_20_21_dist, method = "ward.D2")
plot(as.dendrogram(d1_ward_20_21))
```

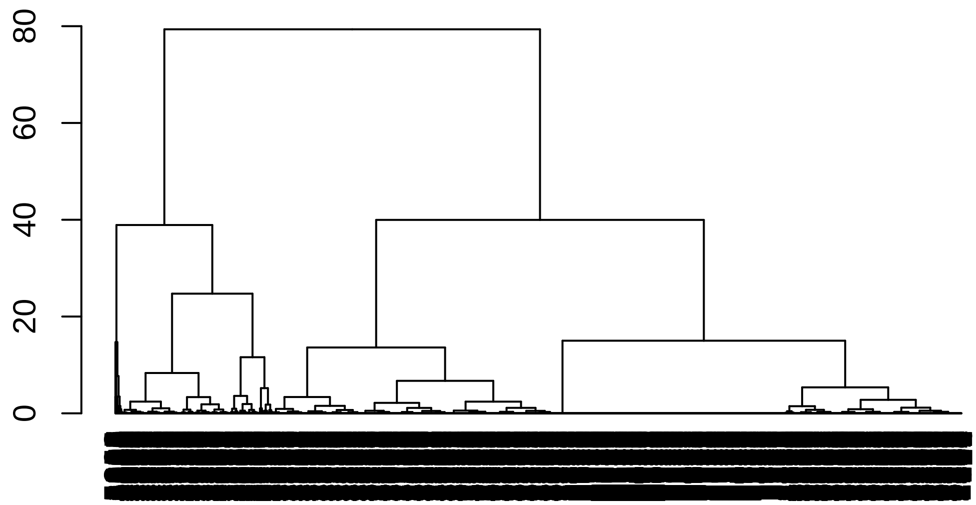

Figure 6: Dendrogram for Ward's method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 2.

```
# Adjusting Ward's method with the chosen numbers of clusters
d1_ward2_20_21_class <- cutree(d1_ward_20_21, k = 2)
d1_ward4_20_21_class <- cutree(d1_ward_20_21, k = 4)
```

4.2.4 Single linkage

4.2.4.1 2018-2019

The dendrogram for the single linkage method for the 2018-2019 period and 10 to 14 age group can be seen in **Figure 7**.

```
# Ploting the dendrogram for the 2018-2019 period
set.seed(1504)
d1_single_18_19 <- hclust(dados1_18_19_dist, method = "single")
plot(as.dendrogram(d1_single_18_19))
```

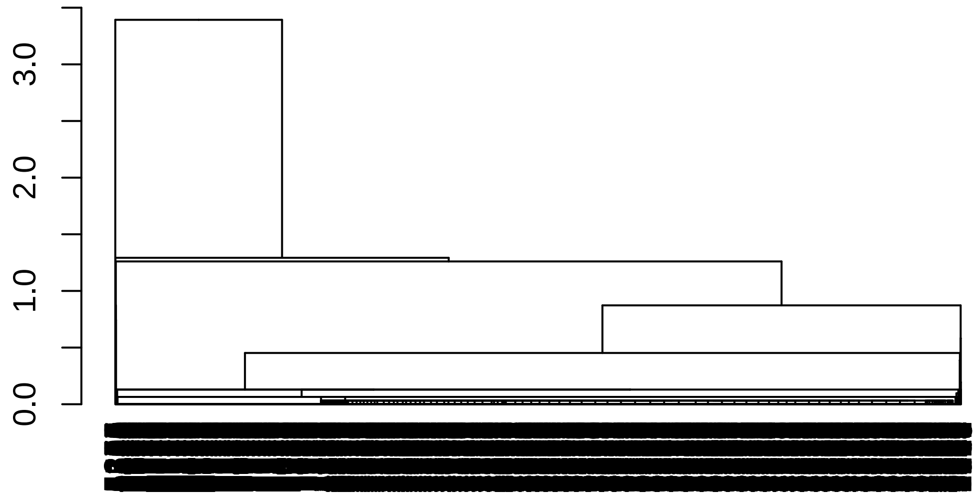

Figure 7: Dendrogram for the single linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.2.4.2 2020-2021

The dendrogram for the single linkage method for the 2020-2021 period and 10 to 14 age group can be seen in **Figure 8**.

```
# Ploting the dendrogram for the 2020-2021 period
set.seed(1504)
d1_single_20_21 <- hclust(dados1_20_21_dist, method = "single")
plot(as.dendrogram(d1_single_20_21))
```

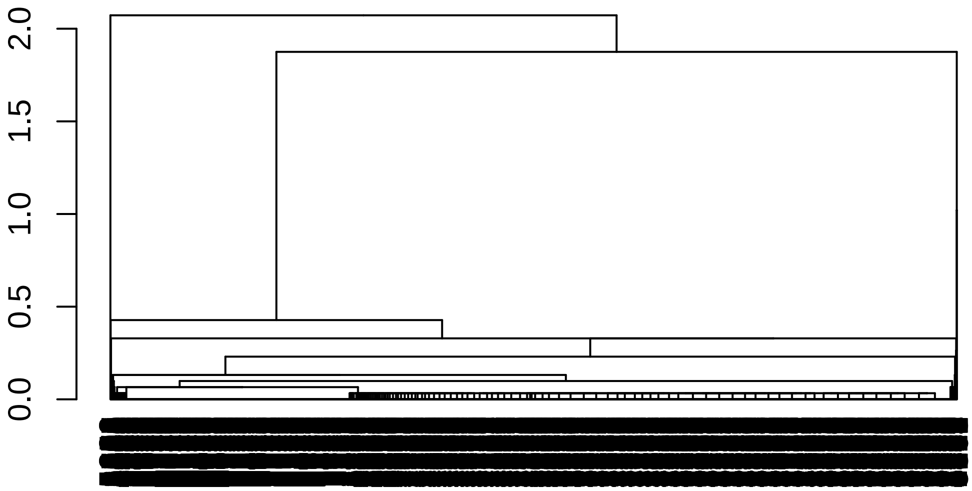

Figure 8: Dendrogram for the single linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.2.5 Complete linkage

4.2.5.1 2018-2019

The dendrogram for the complete linkage method for the 2018-2019 period and 10 to 14 age group can be seen in **Figure 9**.

```
# Ploting the dendrogram for the complete linkage method for the 2018-2019 period
set.seed(1504)
d1_complete_18_19 <- hclust(dados1_18_19_dist, method = "complete")
plot(as.dendrogram(d1_complete_18_19))
```

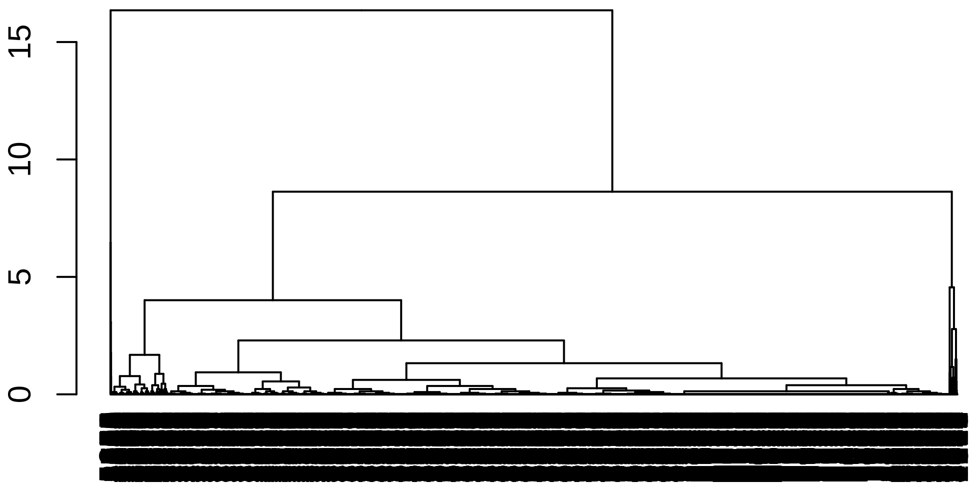

Figure 9: Dendrogram for the complete linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.2.5.2 2020-2021

The dendrogram for the complete linkage method for the 2020-2021 period and 10 to 14 age group can be seen in **Figure 10**.

```
# Ploting the dendrogram for the complete linkage method for the 2020-2021 period
set.seed(1504)
d1_complete_20_21 <- hclust(dados1_20_21_dist, method = "complete")
plot(as.dendrogram(d1_complete_20_21))
```

Figure 10: Dendrogram for the complete linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.2.6 Average linkage

4.2.6.1 2018-2019

The dendrogram for the average linkage method for the 2018-2019 period and 10 to 14 age group can be seen in **Figure 11**.

```
# Ploting the dendrogram for the average linkage method for the 2018-2019 period
set.seed(1504)
d1_average_18_19 <- hclust(dados1_18_19_dist, method = "average")
plot(as.dendrogram(d1_average_18_19))
```

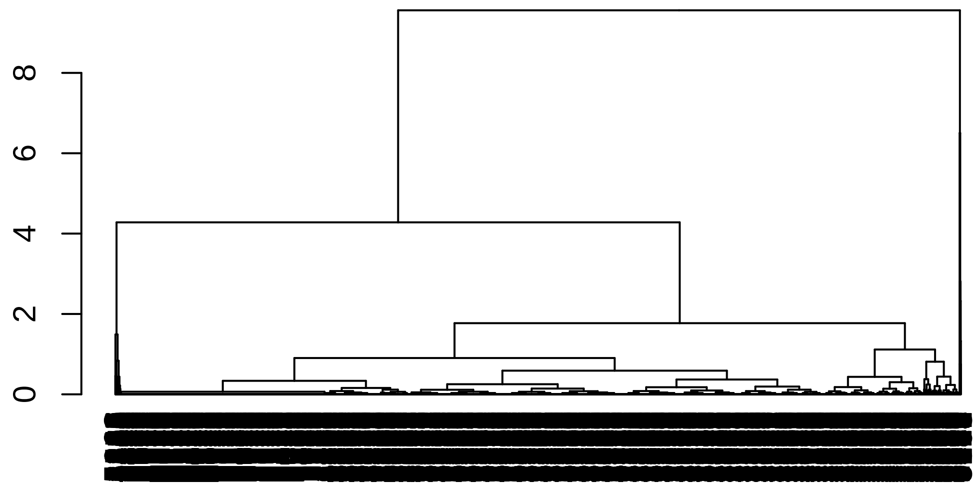

Figure 11: Dendrogram for the average linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.2.6.2 2020-2021

The dendrogram for the average linkage method for the 2020-2021 period and 10 to 14 age group can be seen in **Figure 12**.

```
# Ploting the dendrogram for the average linkage method for the 2020-2021 period
set.seed(1504)
d1_average_20_21 <- hclust(dados1_20_21_dist, method = "average")
plot(as.dendrogram(d1_average_20_21))
```

Figure 12: Dendrogram for the average linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.2.7 Choosing the best clustering method

The code below contains a function that calculates the evaluation indices for each number of clusters and candidate method.

```
# Creating a function that calculates the considered indices
avaliar_cluster_indices <- function(tipo_taxa, anos, kmeans_n, pam_n, ward_n) {
  dados_nome <- paste0("dados", substr(tipo_taxa, 2, 2), "_", anos)
  dist_nome <- paste0("dados", substr(tipo_taxa, 2, 2), "_", anos, "_dist")

  dados <- get(dados_nome)
  dist <- get(dist_nome)

  # K-means
  kmeans_index <- data.frame(
    method = unlist(lapply(kmeans_n, function(i) paste0("kmeans", i))),
    db_index_cent = unlist(lapply(kmeans_n, function(i) {
      index.DB(dados, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster)$DB
    })),
    db_index_med = unlist(lapply(kmeans_n, function(i) {
      index.DB(dados, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster,
               d = dist, centrotypes = "medoids")$DB
    })),
    dunn_index = unlist(lapply(kmeans_n, function(i) {
      dunn(distance = dist, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster)
    })),
```

```r
  silh_index = unlist(lapply(kmeans_n, function(i) {
    index.S(dist, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster)
  })),
  ch_index_cent = unlist(lapply(kmeans_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster)
  })),
  ch_index_med = unlist(lapply(kmeans_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_kmeans", i, "_", anos))$cluster,
             d = dist, centrotypes = "medoids")
  }))
)

# PAM
pam_index <- data.frame(
  method = unlist(lapply(pam_n, function(i) paste0("pam", i))),
  db_index_cent = unlist(lapply(pam_n, function(i) {
    index.DB(dados, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster)$DB
  })),
  db_index_med = unlist(lapply(pam_n, function(i) {
    index.DB(dados, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster,
             d = dist, centrotypes = "medoids")$DB
  })),
  dunn_index = unlist(lapply(pam_n, function(i) {
    dunn(distance = dist, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster)
  })),
  silh_index = unlist(lapply(pam_n, function(i) {
    index.S(dist, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster)
  })),
  ch_index_cent = unlist(lapply(pam_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster)
  })),
  ch_index_med = unlist(lapply(pam_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_pam", i, "_", anos))$cluster,
             d = dist, centrotypes = "medoids")
  }))
)

# Ward
ward_index <- data.frame(
  method = unlist(lapply(ward_n, function(i) paste0("ward", i))),
  db_index_cent = unlist(lapply(ward_n, function(i) {
    index.DB(dados, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")))$DB
  })),
  db_index_med = unlist(lapply(ward_n, function(i) {
    index.DB(dados, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")),
             d = dist, centrotypes = "medoids")$DB
  })),
  dunn_index = unlist(lapply(ward_n, function(i) {
    dunn(distance = dist, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")))
  })),
  silh_index = unlist(lapply(ward_n, function(i) {
    index.S(dist, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")))
  })),
  ch_index_cent = unlist(lapply(ward_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")))
  })),
  ch_index_med = unlist(lapply(ward_n, function(i) {
    index.G1(dados, get(paste0(tipo_taxa, "_ward", i, "_", anos, "_class")),
```

```
              d = dist, centrotypes = "medoids")
  }))
 )

 avaliacao <- rbind(pam_index, kmeans_index, ward_index)
 colnames(avaliacao) <- c(
   "Method", "Davies-Bouldin (centroid)", "Davies-Bouldin (medoid)",
   "Dunn index", "Silhouette index", "Calinski-Harabasz (centroid)",
   "Calinski-Harabasz (medoid)"
 )
 return(avaliacao)
}
```

4.2.7.1 2018-2019

The evaluation indices of the candidate clusterings for the 2018-2019 period and 10 to 14 age group can be seen in **Table 5**.

```
# Calculating the considered indices
d1_avaliacao_18_19 <- avaliar_cluster_indices(
  tipo_taxa = "d1",
  anos = "18_19",
  kmeans_n = 3:4,
  pam_n = 3:4,
  ward_n = 3
)

d1_avaliacao_18_19 |>
  kable(
    caption = "Evaluation indices of the candidate clusterings for the 2018-2019
    period and 10 to 14 age group.",
    align = "cccccc"
  ) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(2:7, width = "2cm")
```

Table 5: Evaluation indices of the candidate clusterings for the 2018-2019 period and 10 to 14 age group.

Method	Davies-Bouldin (centroid)	Davies-Bouldin (medoid)	Dunn index	Silhouette index	Calinski-Harabasz (centroid)	Calinski-Harabasz (medoid)
pam3	0.8381454	1.0445179	0.0021692	0.5637583	5815.227	5531.958
pam4	0.8227825	1.0714429	0.0022371	0.5868078	5997.481	5694.880
kmeans3	0.8412399	1.0101635	0.0023256	0.5807404	7689.521	7491.191
kmeans4	0.7130065	0.9078588	0.0027100	0.5806522	9205.851	7830.910
ward3	0.8452240	1.0319135	0.0023641	0.5846469	7627.190	7394.749

Comments:

- **Davies Bouldin (centroid):** the lower the better (kmeans4, followed by kmedoids4);
- **Davies Bouldin (medoid):** the lower the better (kmeans4, followed by kmeans3 and ward3);
- **Dunn Index:** the higher the better (kmeans4, followed by ward3 and kmeans3);
- **Silhouette Index:** the higher the better (kmedoids4, followed by ward3, kmeans3 and kmeans4);

- **Calinski-Harabasz (centroid):** the higher the better (kmeans4, followed by kmeans3 and ward3);
- **Calinski-Harabasz (medoid):** the higher the better (kmeans4, followed by kmeans3 and ward3).

Choice: despite the better performance of K-Means with K = 4, one of its clusters contained only a few municipalities (46).

```
d1_kmeans4_18_19$size
```

```
## [1]  845 2366   46 2313
```

Thus, the chosen method was the **K-Means with K = 3**, since it had the best average performance after K-Means with K = 4.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_18_19$cluster_10_a_14 <- d1_kmeans3_18_19$cluster

# Ordering by mean fertility rate
ordem_taxa1_18_19 <- df_indicadores_18_19 |>
  group_by(cluster_10_a_14) |>
  summarise(tx_fecundidade_10_a_14 = mean(tx_fecundidade_10_a_14)) |>
  ungroup() |>
  arrange(tx_fecundidade_10_a_14) |>
  pull(cluster_10_a_14)

df_indicadores_18_19$cluster_10_a_14 <- factor(case_when(
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[1] ~
    "1: lowest fertility rates",
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[2] ~
    "2: intermediary fertility rates",
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[3] ~
    "3: highest fertility rates"
))
```

4.2.7.2 2020-2021

The evaluation indices of the candidate clusterings for the 2020-2021 period and 10 to 14 age group can be seen in **Table 6**.

```
# Calculating the considered indices
d1_avaliacao_20_21 <- avaliar_cluster_indices(
  tipo_taxa = "d1",
  anos = "20_21",
  kmeans_n = 3:4,
  pam_n = 3:4,
  ward_n = 2
)

d1_avaliacao_20_21 |>
  kable(
    caption = "Evaluation indices of the candidate clusterings for the 2020-2021
    period and 10 to 14 age group.",
    align = "cccccc"
  ) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(2:7, width = "2cm")
```

Table 6: Evaluation indices of the candidate clusterings for the 2020-2021 period and 10 to 14 age group.

Method	Davies-Bouldin (centroid)	Davies-Bouldin (medoid)	Dunn index	Silhouette index	Calinski-Harabasz (centroid)	Calinski-Harabasz (medoid)
pam3	0.8324863	1.0948785	0.0027174	0.5618952	5804.152	5354.868
pam4	0.7781750	1.0464116	0.0028818	0.5942662	7199.515	6770.703
kmeans3	0.7657205	0.9256175	0.0030395	0.5980689	8927.850	8675.012
kmeans4	0.6632878	0.7294509	0.0034014	0.5977349	11364.869	10444.413
ward2	0.8429884	1.0480163	0.0027701	0.6401414	7243.426	7022.450

Comments:

- **Davies Bouldin (centroid):** the lower the better (kmeans4, followed by kmeans3);
- **Davies Bouldin (medoid):** the lower the better (kmeans4, followed by kmeans3);
- **Dunn Index:** the higher the better (kmeans4, followed by kmeans3);
- **Silhouette Index:** the higher the better (ward2, followed by kmeans3);
- **Calinski-Harabasz (centroid):** the higher the better (kmeans4, followed by kmeans3);
- **Calinski-Harabasz (medoid):** the higher the better (kmeans4, followed by kmeans3).

Choice: again, despite the better performance of K-means with K = 4, one of its clusters contained only a few municipalities (93).

```
d1_kmeans4_20_21$size
```

```
## [1] 2107 2459   93  911
```

Thus, the chosen method was the **K-Means with K = 3**, since it had the best overall performance after K-means with K = 4.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_20_21$cluster_10_a_14 <- d1_kmeans3_20_21$cluster

# Ordering by mean fertility rate
ordem_taxa1_20_21 <- df_indicadores_20_21 |>
  group_by(cluster_10_a_14) |>
  summarise(tx_fecundidade_10_a_14 = mean(tx_fecundidade_10_a_14)) |>
  ungroup() |>
  arrange(tx_fecundidade_10_a_14) |>
  pull(cluster_10_a_14)

df_indicadores_20_21$cluster_10_a_14 <- factor(case_when(
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[1] ~
    "1: lowest fertility rates",
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[2] ~
    "2: intermediary fertility rates",
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[3] ~
    "3: highest fertility rates"
))
```

4.3 For the 15 to 19 age group

4.3.1 K-Means

4.3.1.1 2018-2019

The K-Means knee plot for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 13**.

```
# Ploting the k-means knee-plot for the 2018-2019 period
set.seed(1504)
fviz_nbclust(dados2_18_19, kmeans, method = "wss")
```

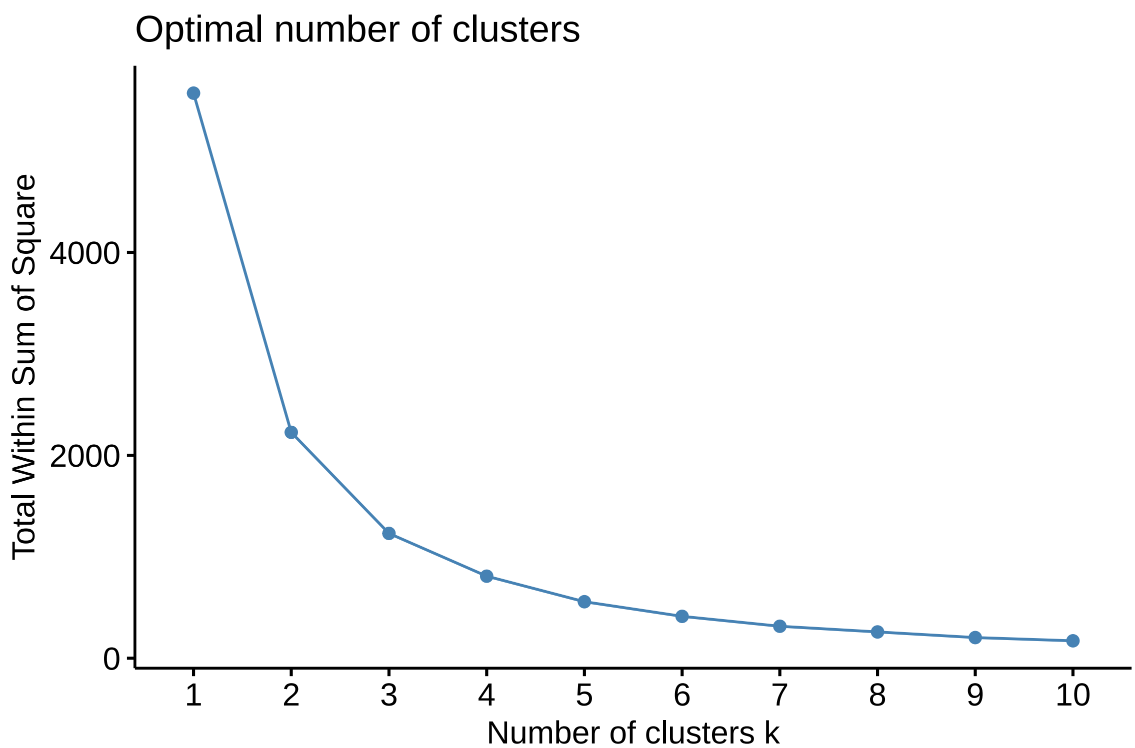

Figure 13: K-Means knee-plot for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3 (an argument can be made that only after this value the total within-group variance does not decrease significantly).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d2_kmeans3_18_19 <- kmeans(dados2_18_19, 3)
```

4.3.1.2 2020-2021

The K-Means knee plot for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 14**.

```
# Ploting the k-means knee-plot for the 2020-2021 period
set.seed(1504)
fviz_nbclust(dados2_20_21, kmeans, method = "wss")
```

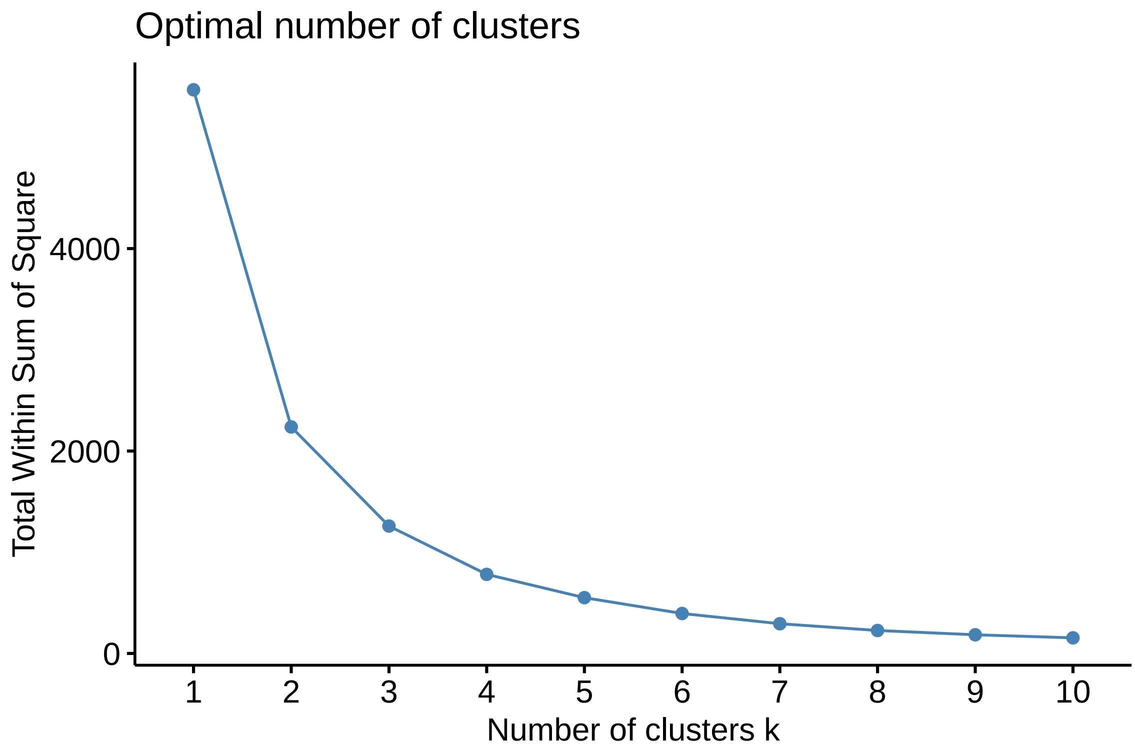

Figure 14: K-Means knee-plot for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 3 (an argument can be made that only after this value the total within-group variance does not decrease significantly).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d2_kmeans3_20_21 <- kmeans(dados2_20_21, 3)
```

4.3.2 K-Medians

4.3.2.1 2018-2019

The K-Medians knee plot for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 15**.

```
# Ploting the k-medoids knee-plot for the 2018-2019 period
set.seed(1504)
fviz_nbclust(dados2_18_19, cluster::pam, method = "wss")
```

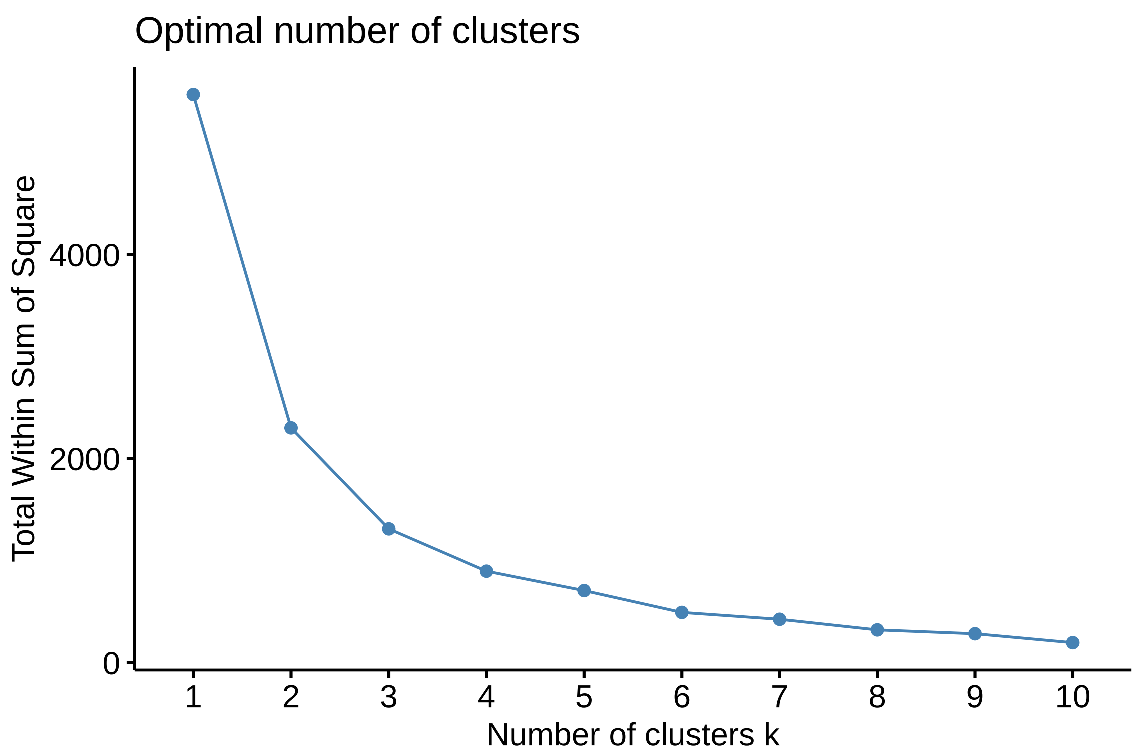

Figure 15: K-Medians knee-plot for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3 (an argument can be made that only after this value the total within-group variance does not decrease significantly).

```r
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d2_kmedoids3_18_19 <- cluster::pam(dados2_18_19, k = 3)
```

4.3.2.2 2020-2021

The K-Medians knee plot for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 16**.

```r
# Ploting the k-medoids knee-plot for the 2020-2021 period
set.seed(1504)
fviz_nbclust(dados2_20_21, cluster::pam, method = "wss")
```

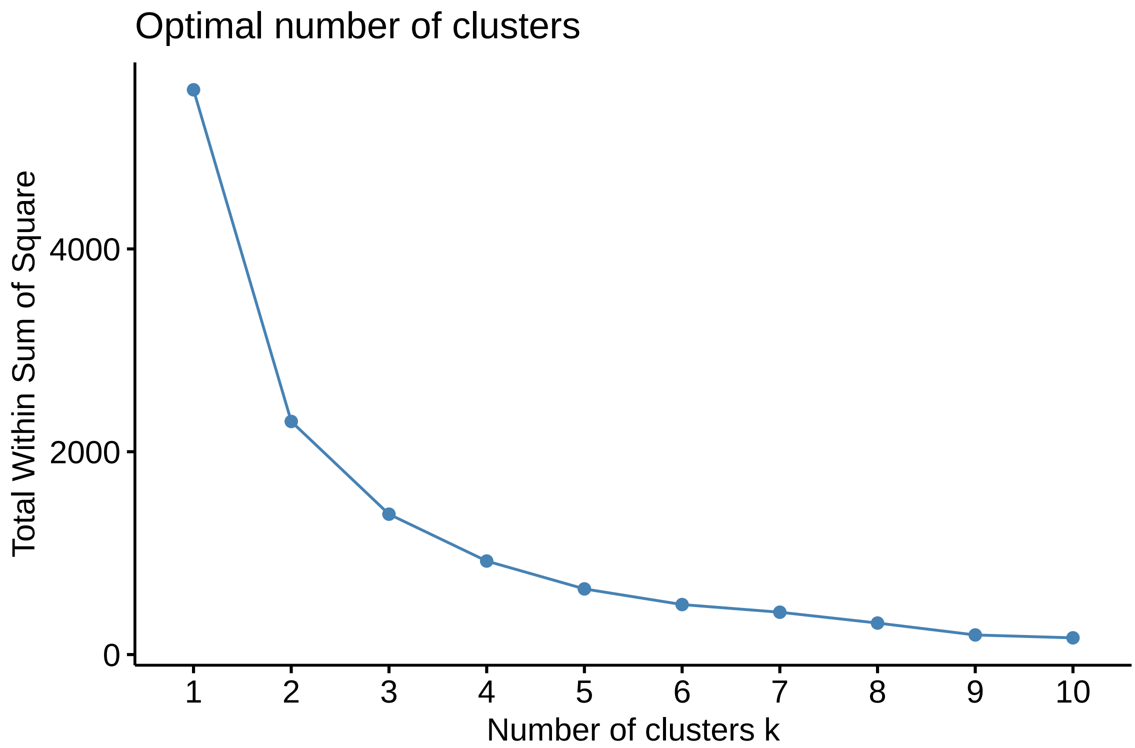

Figure 16: K-Medians knee-plot for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 3 (an argument can be made that only after this value the total within-group variance does not decrease significantly).

```
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d2_kmedoids3_20_21 <- cluster::pam(dados2_20_21, k = 3)
```

4.3.3 Ward's method

4.3.3.1 2018-2019

The dendrogram for Ward's method for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 17**.

```
# Ploting the drodrogram for Ward's method for the 2018-2019 period
set.seed(1504)
d2_ward_18_19 <- hclust(dados2_18_19_dist, method = "ward.D2")
plot(as.dendrogram(d2_ward_18_19))
```

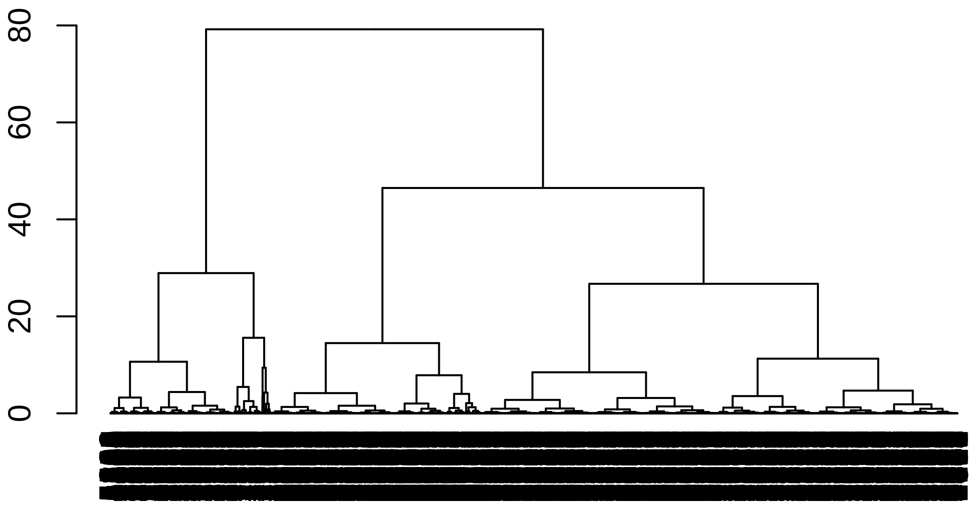

Figure 17: Dendrogram for Ward's method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3.

```
# Adjusting Ward's method with the chosen numbers of clusters
d2_ward3_18_19_class <- cutree(d2_ward_18_19, k = 3)
```

4.3.3.2 2020-2021

The dendrogram for Ward's method for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 18**.

```
# Ploting the drodrogram for Ward's method for the 2020-2021 period
set.seed(1504)
d2_ward_20_21 <- hclust(dados2_20_21_dist, method = "ward.D2")
plot(as.dendrogram(d2_ward_20_21))
```

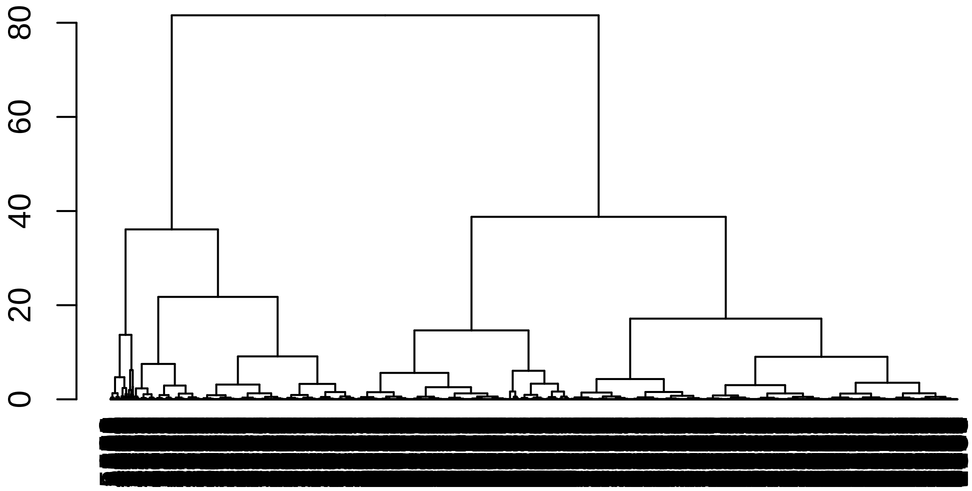

Figure 18: Dendrogram for Ward's method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 2 and K = 3.

```
# Adjusting Ward's method with the chosen numbers of clusters
d2_ward2_20_21_class <- cutree(d2_ward_20_21, k = 2)
d2_ward3_20_21_class <- cutree(d2_ward_20_21, k = 3)
```

4.3.4 Single linkage

4.3.4.1 2018-2019

The dendrogram for the single linkage method for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 19**.

```
# Ploting the drodrogram for the single linkage method for the 2018-2019 period
set.seed(1504)
d2_single_18_19 <- hclust(dados2_18_19_dist, method = "single")
plot(as.dendrogram(d2_single_18_19))
```

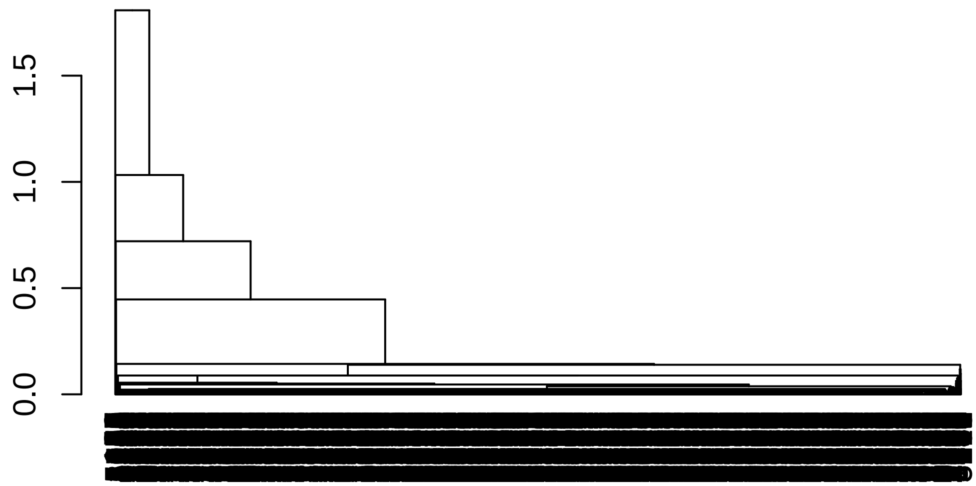

Figure 19: Dendrogram for the single linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.3.4.2 2020-2021

The dendrogram for the single linkage method for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 20**.

```
# Ploting the drodrogram for the single linkage method for the 2020-2021 period
set.seed(1504)
d2_single_20_21 <- hclust(dados2_20_21_dist, method = "single")
plot(as.dendrogram(d2_single_20_21))
```

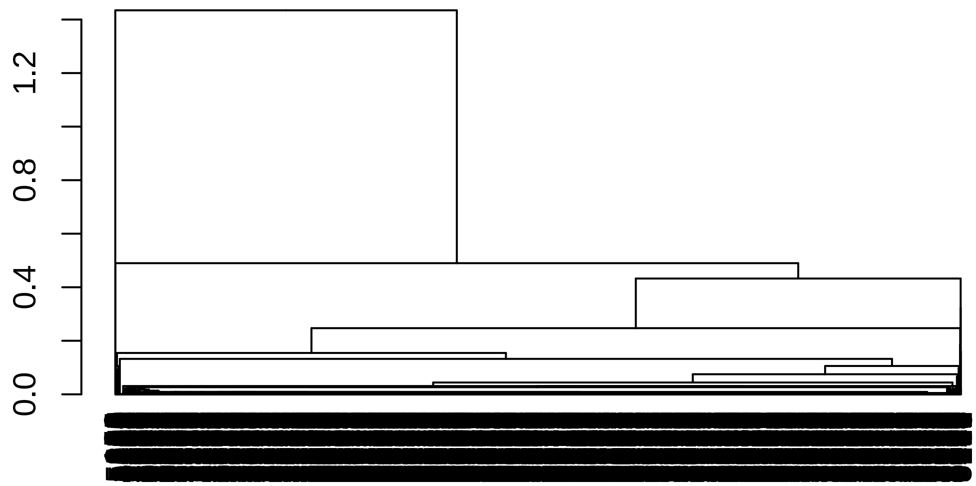

Figure 20: Dendrogram for the single linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.3.5 Complete linkage

4.3.5.1 2018-2019

The dendrogram for the complete linkage method for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 21**.

```
# Ploting the drodrogram for the complete linkage method for the 2018-2019 period
set.seed(1504)
d2_complete_18_19 <- hclust(dados2_18_19_dist, method = "complete")
plot(as.dendrogram(d2_complete_18_19))
```

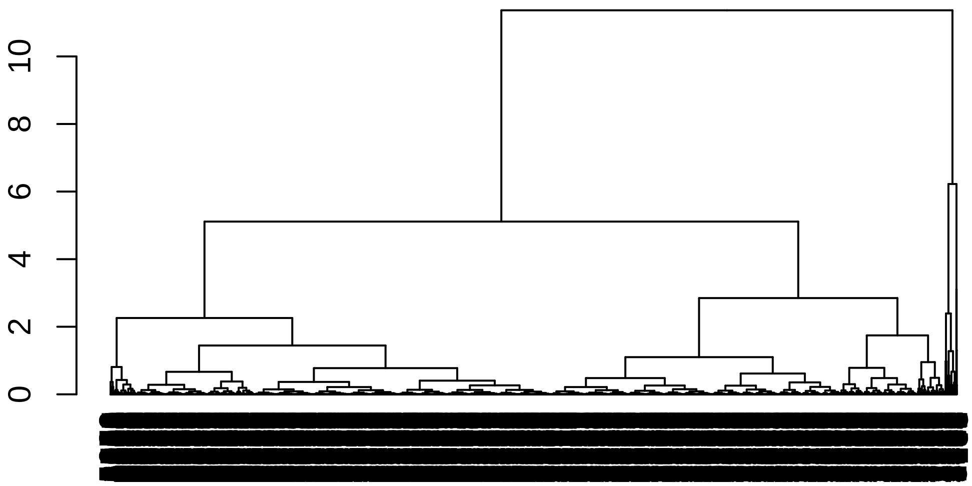

Figure 21: Dendrogram for the complete linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.3.5.2 2020-2021

The dendrogram for the complete linkage method for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 22**.

```
# Ploting the drodrogram for the complete linkage method for the 2020-2021 period
set.seed(1504)
d2_complete_20_21 <- hclust(dados2_20_21_dist, method = "complete")
plot(as.dendrogram(d2_complete_20_21))
```

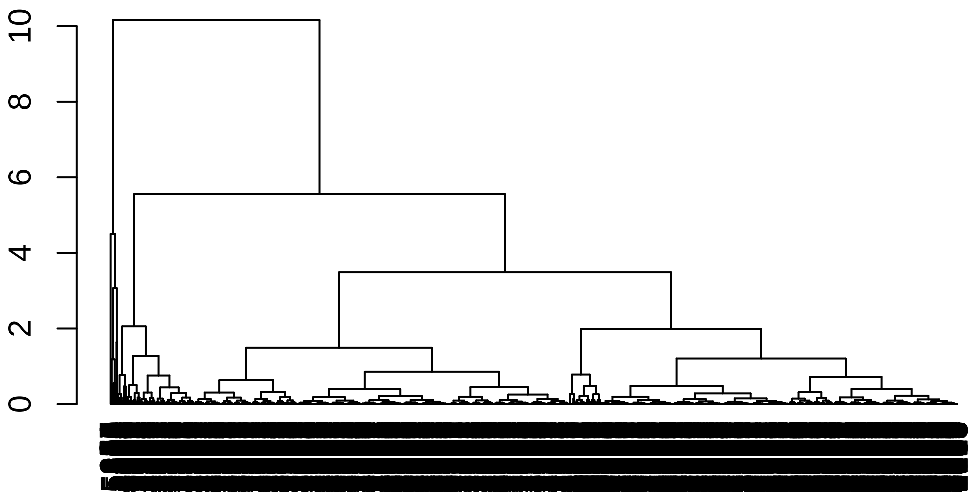

Figure 22: Dendrogram for the complete linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.3.6 Average linkage

4.3.6.1 2018-2019

The dendrogram for the average linkage method for the 2018-2019 period and 15 to 19 age group can be seen in **Figure 23**.

```r
# Ploting the drodrogram for the average linkage method for the 2018-2019 period
set.seed(1504)
d2_average_18_19 <- hclust(dados2_18_19_dist, method = "average")
plot(as.dendrogram(d2_average_18_19))
```

Figure 23: Dendrogram for the average linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

4.3.6.2 2020-2021

The dendrogram for the average linkage method for the 2020-2021 period and 15 to 19 age group can be seen in **Figure 24**.

```
# Ploting the drodrogram for the average linkage method for the 2020-2021 period
set.seed(1504)
d2_average_20_21 <- hclust(dados2_20_21_dist, method = "average")
plot(as.dendrogram(d2_average_20_21))
```

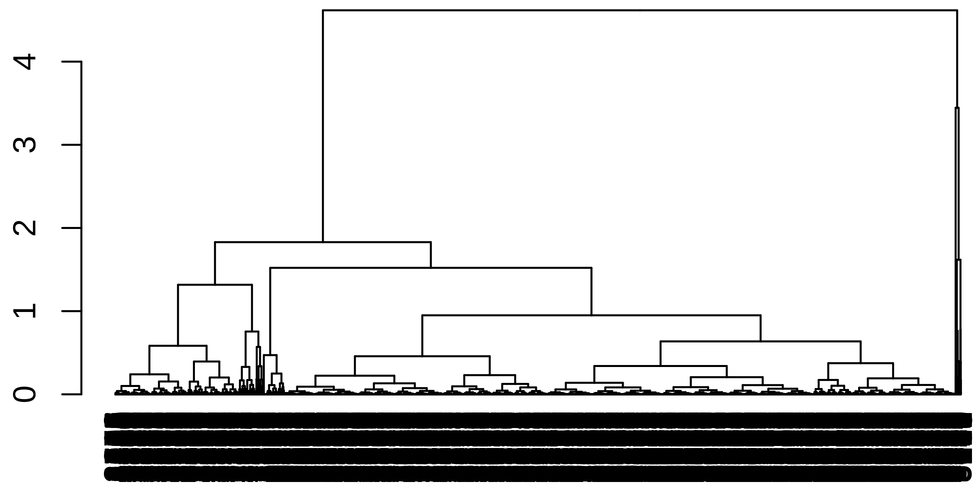

Figure 24: Dendrogram for the average linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

4.3.7 Choosing the best clustering method

4.3.7.1 2018-2019

The evaluation indices of the candidate clusterings for the 2018-2019 period and 15 to 19 age group can be seen in **Table 7**.

```
# Calculating the considered indices
d2_avaliacao_18_19 <- avaliar_cluster_indices(
  tipo_taxa = "d2",
  anos = "18_19",
  kmeans_n = 3,
  pam_n = 3,
  ward_n = 3
)

d2_avaliacao_18_19 |>
  kable(
    caption = "Evaluation indices of the candidate clusterings for the 2018-2019
    period and 15 to 19 age group.",
    align = "cccccc"
  ) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(2:7, width = "2cm")
```

Table 7: Evaluation indices of the candidate clusterings for the 2018-2019 period and 15 to 19 age group.

Method	Davies-Bouldin (centroid)	Davies-Bouldin (medoid)	Dunn index	Silhouette index	Calinski-Harabasz (centroid)	Calinski-Harabasz (medoid)
pam3	0.7707887	0.9133970	0.0005025	0.5160113	9035.924	8576.450
kmeans3	0.7433856	0.8754579	0.0005283	0.5374209	9821.335	9387.630
ward3	0.7395803	0.8571475	0.0005112	0.5008676	8675.739	8352.971

Comments:

- **Davies Bouldin (centroid):** the lower the better (ward3, followed by kmeans3);
- **Davies Bouldin (medoid):** the lower the better (ward3, followed by kmeans3);
- **Dunn Index:** the higher the better (kmeans3, followed by ward3);
- **Silhouette Index:** the higher the better (kmeans3, followed by kmedoids3);
- **Calinski-Harabasz (centroid):** the higher the better (kmeans3, followed by kmedoids3);
- **Calinski-Harabasz (medoid):** the higher the better (kmeans3, followed by kmedoids3).

Choice: K-Means with K = 3, since it performed better in most of the metrics considered.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_18_19$cluster_15_a_19 <- d2_kmeans3_18_19$cluster

# Ordering by mean fertility rate
ordem_taxa2_18_19 <- df_indicadores_18_19 |>
  group_by(cluster_15_a_19) |>
  summarise(tx_fecundidade_15_a_19 = mean(tx_fecundidade_15_a_19)) |>
  ungroup() |>
  arrange(tx_fecundidade_15_a_19) |>
  pull(cluster_15_a_19)

df_indicadores_18_19$cluster_15_a_19 <- factor(case_when(
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[1] ~
    "1: lowest fertility rates",
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[2] ~
    "2: intermediary fertility rates",
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[3] ~
    "3: highest fertility rates"
))
```

4.3.7.2 2020-2021

The evaluation indices of the candidate clusterings for the 2020-2021 period and 15 to 19 age group can be seen in **Table 8**.

```
# Calculating the considered indices
d2_avaliacao_20_21 <- avaliar_cluster_indices(
  tipo_taxa = "d2",
  anos = "20_21",
  kmeans_n = 3,
  pam_n = 3,
  ward_n = 2:3
)

d2_avaliacao_20_21 |>
```

```r
  kable(
    caption = "Evaluation indices of the candidate clusterings for the 2020-2021
    period and 15 to 19 age group.",
    align = "cccccc"
  ) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE) |>
  column_spec(2:7, width = "2cm")
```

Table 8: Evaluation indices of the candidate clusterings for the 2020-2021 period and 15 to 19 age group.

Method	Davies-Bouldin (centroid)	Davies-Bouldin (medoid)	Dunn index	Silhouette index	Calinski-Harabasz (centroid)	Calinski-Harabasz (medoid)
pam3	0.7759754	0.9472968	0.0005963	0.5079739	8415.051	7844.758
kmeans3	0.7479520	0.8919405	0.0006464	0.5303942	9533.028	9058.344
ward2	0.8014165	0.9982053	0.0005838	0.5808038	8269.139	7895.314
ward3	0.7703716	0.9286440	0.0005838	0.5015726	7624.941	7133.159

Comments:

- **Davies Bouldin (centroid):** the lower the better (kmeans3, followed by ward3);
- **Davies Bouldin (medoid):** the lower the better (kmeans3, followed by ward3);
- **Dunn Index:** the higher the better (kmeans3, followed by kmedoids3);
- **Silhouette Index:** the higher the better (ward2, followed by kmeans3);
- **Calinski-Harabasz (centroid):** the higher the better (kmeans3, followed by kmedoids3);
- **Calinski-Harabasz (medoid):** the higher the better (kmeans3, followed by ward2).

Choice: K-Means with K = 3, since it performed better in most of the metrics considered.

```r
# Adding the column with the clusters information to the original dataframe
df_indicadores_20_21$cluster_15_a_19 <- d2_kmeans3_20_21$cluster

# Ordering by mean fertility rate
ordem_taxa2_20_21 <- df_indicadores_20_21 |>
  group_by(cluster_15_a_19) |>
  summarise(tx_fecundidade_15_a_19 = mean(tx_fecundidade_15_a_19)) |>
  ungroup() |>
  arrange(tx_fecundidade_15_a_19) |>
  pull(cluster_15_a_19)

df_indicadores_20_21$cluster_15_a_19 <- factor(case_when(
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[1] ~
    "1: lowest fertility rates",
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[2] ~
    "2: intermediary fertility rates",
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[3] ~
    "3: highest fertility rates"
))
```

4.4 Visualizing the clusters in a map

```r
# Downloading the geometry data
df_muni_sf <- read_municipality(year = 2019, showProgress = FALSE) |>
  mutate(codmunres = as.numeric(substr(code_muni, 1, 6)))

df_ufs_sf <- read_state(year = 2019, showProgress = FALSE)

# Joining the geometry and the clustering data for each period
df_dados_mapa_18_19 <- left_join(df_indicadores_18_19, df_muni_sf) |>
  mutate(periodo = "2018-2019")

df_dados_mapa_20_21 <- left_join(df_indicadores_20_21, df_muni_sf) |>
  mutate(periodo = "2020-2021")

df_dados_mapa_completo <- full_join(df_dados_mapa_18_19, df_dados_mapa_20_21) |>
  st_as_sf()
```

For the 10 to 14 years age group, group 3 (municipalities with the highest fertility rates) is predominantly formed by municipalities in the North region of Brazil, while group 1 (municipalities with the lowest fertility rates) is mainly composed of municipalities in the Southeast and South regions, with a similar pattern for the two periods of analysis (**Figure 25**).

```r
# Plotting the map for the 10 to 14 age group
ggplot() +
  geom_sf(
    data = df_dados_mapa_completo,
    aes(fill = cluster_10_a_14),
    color = NA
  ) +
  facet_wrap(vars(periodo)) +
  scale_fill_viridis_d(
    name = "Groups",
    end = 0.8,
    alpha = 0.6,
    direction = -1
  ) +
  geom_sf(data = df_ufs_sf, fill = NA, linewidth = 0.08, color = "black") +
  theme_bw() +
  theme(legend.position = "bottom")
```

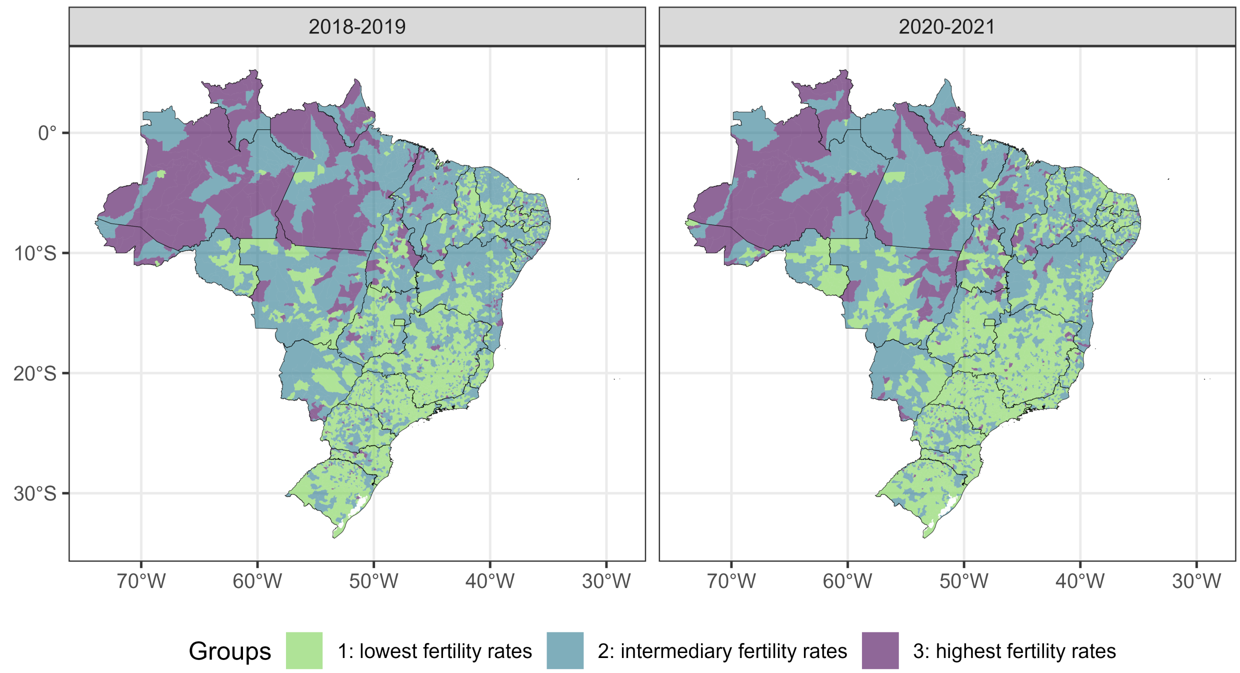

Figure 25: Distribution of groups of municipalities formed by municipalities with similar fertility rate of women aged 10 to 14 years in the pre-pandemic period (on the left) and in the pandemic period (on the right).

For the 15 to 19 years old age group, group 3 (municipalities with the highest fertility rates) is predominantly formed by municipalities located in the North of Brazil and in the Southeast of the state of Mato Grosso. Group 1 (municipalities with the lowest fertility rates) is predominantly composed of municipalities in the Southeast and South regions (**Figure 26**).

```
# Plotting the map for the 15 to 19 age group
ggplot() +
  geom_sf(
    data = df_dados_mapa_completo,
    aes(fill = cluster_15_a_19),
    color = NA
  ) +
  facet_wrap(vars(periodo)) +
  scale_fill_viridis_d(
    name = "Groups",
    end = 0.8,
    alpha = 0.6,
    direction = -1
  ) +
  geom_sf(data = df_ufs_sf, fill = NA, linewidth = 0.08, color = "black") +
  theme_bw() +
  theme(legend.position = "bottom")
```

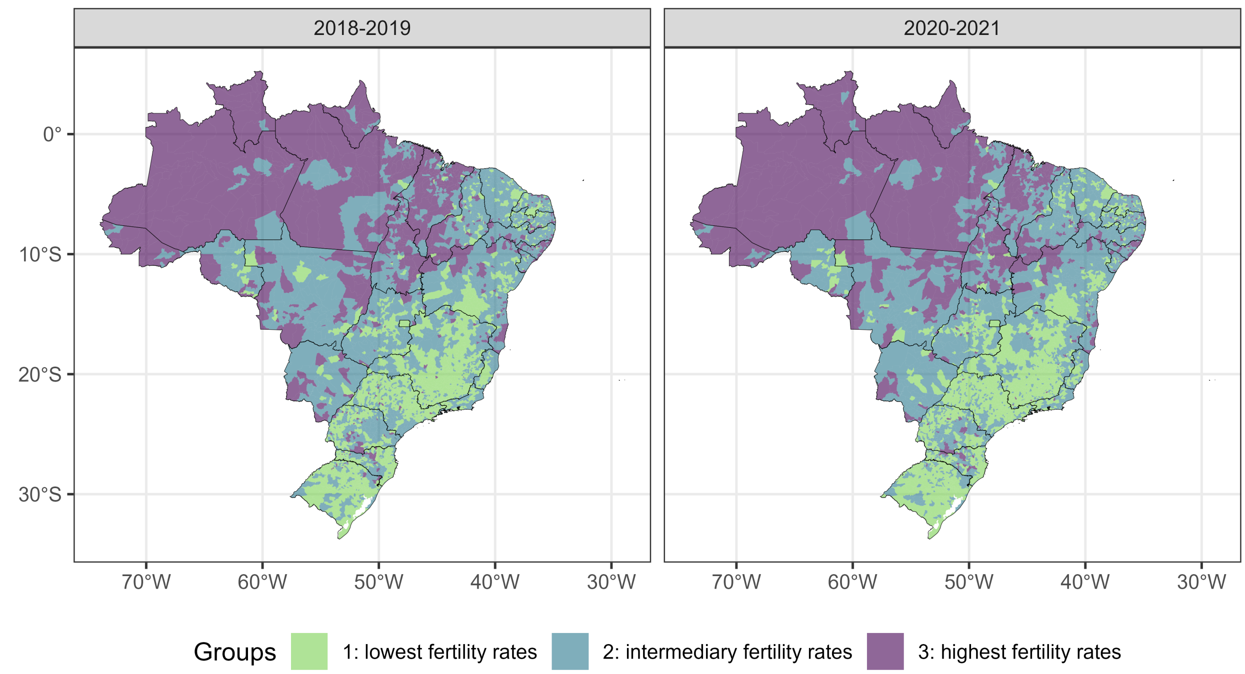

Figure 26: Distribution of groups of municipalities formed by municipalities with similar fertility rate of women aged 15 to 19 years in the pre-pandemic period (on the left) and in the pandemic period (on the right).

4.5 Summary tables

The code below contains a function that creates tables with the summary measures of each group for each variable of interest.

```
# Creating a function that creates a table with the summary measures
cria_tabelas <- function(
  variaveis,
  nomes_legiveis,
  df,
  var_grupos,
  label_tabela,
  faixa_etaria,
  periodo
) {
  grupos <- levels(df[[var_grupos]])

  for (variavel in variaveis) {
    tabela <- data.frame(
      grupo = grupos,
      n = unlist(lapply(
        grupos,
        function(grupo) df[df[[var_grupos]] == grupo, ] |> nrow()
      )),
      minimo = round(
        unlist(lapply(
          grupos,
          function(grupo)
            df[df[[var_grupos]] == grupo, ] |>
```

```
        pull(variavel) |>
        min(na.rm = TRUE)
  )),
  2
),
primeiro_qt = round(
  unlist(lapply(
    grupos,
    function(grupo)
      df[df[[var_grupos]] == grupo, ] |>
        pull(variavel) |>
        quantile(0.25, na.rm = TRUE)
  )),
  2
),
media = round(
  unlist(lapply(
    grupos,
    function(grupo)
      df[df[[var_grupos]] == grupo, ] |>
        pull(variavel) |>
        mean(na.rm = TRUE)
  )),
  2
),
mediana = round(
  unlist(lapply(
    grupos,
    function(grupo)
      df[df[[var_grupos]] == grupo, ] |>
        pull(variavel) |>
        median(na.rm = TRUE)
  )),
  2
),
dp = round(
  unlist(lapply(
    grupos,
    function(grupo)
      df[df[[var_grupos]] == grupo, ] |>
        pull(variavel) |>
        sd(na.rm = TRUE)
  )),
  2
),
terceiro_qt = round(
  unlist(lapply(
    grupos,
    function(grupo)
      df[df[[var_grupos]] == grupo, ] |>
        pull(variavel) |>
        quantile(0.75, na.rm = TRUE)
  )),
  2
),
maximo = round(
  unlist(lapply(
    grupos,
```

```r
      function(grupo)
        df[df[[var_grupos]] == grupo, ] |>
          pull(variavel) |>
          max(na.rm = TRUE)
    )),
    2
  )
)

if (variavel == variaveis[1]) {
  print(
    kable(
      tabela,
      align = "cccccccc",
      col.names = c(
        "Group (cluster)",
        "n",
        "Min.",
        "1st Quartile",
        "Mean",
        "Median",
        "S.D.",
        "3rd Quartile",
        "Max."
      ),
      caption = glue::glue(
        "Summary measures of the variables of interest for the clusters of
        municipalities grouped by the fertility rate of women aged
        {faixa_etaria} (per thousand) in the period of {periodo}.
        \\\\ \\\\ \\textbf{{{nomes_legiveis[[variavel]]}}}"
      ),
      label = paste0("tabela", label_tabela)
    ) |>
      kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
      row_spec(1, bold = TRUE)
  )
} else {
  cat(knitr::asis_output(paste0("**", nomes_legiveis[[variavel]], "**\n\n")))
  print(
    kable(
      tabela,
      align = "cccccccc",
      col.names = c(
        "Group (cluster)",
        "n",
        "Min.",
        "1st Quartile",
        "Mean",
        "Median",
        "S.D.",
        "3rd Quartile",
        "Max."
      )
    ) |>
      kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
      row_spec(0, bold = TRUE)
  )
}
```

```
  }
}
nomes_variaveis1 <- c(
  "tx_fecundidade_10_a_14",
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus"
)
nomes_legiveis1 <- c(
  "Fertility rate (10 to 14 age group)",
  "HDI-M",
  "Primary Care coverage",
  "Percentage of women who are exclusive users of the SUS"
)
names(nomes_legiveis1) <- nomes_variaveis1


nomes_variaveis2 <- c(
  "tx_fecundidade_15_a_19",
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus"
)
nomes_legiveis2 <- c(
  "Fertility rate (15 to 19 age group)",
  "HDI-M",
  "Primary Care coverage",
  "Percentage of women who are exclusive users of the SUS"
)
names(nomes_legiveis2) <- nomes_variaveis2
```

4.5.1 For the 10 to 14 age group

4.5.1.1 2018-2019

```
# Creating the summary tables for the 10 to 14 age group for the 2018-2019 period
cria_tabelas(
  df = df_indicadores_18_19,
  var_grupos = "cluster_10_a_14",
  label_tabela = "1",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  nomes_legiveis = nomes_legiveis1,
  periodo = "2018 to 2019"
)
```

Table 9: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 10 to 14 (per thousand) in the period of 2018 to 2019.

Fertility rate (10 to 14 age group)

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3009	0.0	0.0	0.94	0.9	0.95	1.8	2.6
2: intermediary fertility rates	2192	2.7	3.3	4.44	4.2	1.30	5.4	7.5
3: highest fertility rates	369	7.6	8.3	10.70	9.4	4.59	11.3	50.6

HDI-M

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3009	0.48	0.64	0.68	0.70	0.07	0.74	0.86
2: intermediary fertility rates	2192	0.42	0.58	0.63	0.63	0.06	0.69	0.77
3: highest fertility rates	369	0.44	0.56	0.61	0.60	0.07	0.66	0.76

Primary Care coverage

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3009	0	90.8	91.46	99.8	16.05	100	100
2: intermediary fertility rates	2192	0	94.0	93.42	100.0	13.29	100	100
3: highest fertility rates	369	25	93.0	93.81	100.0	12.32	100	100

Percentage of women who are exclusive users of the SUS

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3009	0	2.3	11.94	7.4	12.35	18.2	84.3
2: intermediary fertility rates	2192	0	1.0	6.12	2.6	8.60	7.7	98.5
3: highest fertility rates	369	0	0.4	2.94	1.2	4.29	3.5	24.8

4.5.1.2 2020-2021

Summary measures of each group for the 2020-2021 period and 10 to 14 age group can be seen in **Table 10**.

```
# Creating the summary tables for the 10 to 14 age group for the 2020-2021 period
cria_tabelas(
  df = df_indicadores_20_21,
  var_grupos = "cluster_10_a_14",
  label_tabela = "2",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  nomes_legiveis = nomes_legiveis1,
  periodo = "2020 to 2021"
)
```

Table 10: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 10 to 14 (per thousand) in the period of 2020 to 2021.

Fertility rate (10 to 14 age group)

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3273	0.0	0.0	0.90	0.8	0.93	1.7	2.6
2: intermediary fertility rates	2008	2.7	3.3	4.48	4.2	1.36	5.4	7.9
3: highest fertility rates	289	8.0	8.7	11.48	10.0	4.45	12.9	40.9

HDI-M

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3273	0.45	0.63	0.68	0.70	0.07	0.73	0.86
2: intermediary fertility rates	2008	0.45	0.58	0.63	0.62	0.06	0.68	0.81
3: highest fertility rates	289	0.42	0.55	0.60	0.59	0.07	0.66	0.76

Primary Care coverage

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3273	21.6	89.9	91.77	100	14.69	100	100
2: intermediary fertility rates	2008	30.0	96.4	95.31	100	10.12	100	100
3: highest fertility rates	289	43.5	95.4	94.91	100	10.41	100	100

Percentage of women who are exclusive users of the SUS

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	3273	0	2.5	12.42	8.1	12.28	18.9	76.2
2: intermediary fertility rates	2008	0	0.9	5.74	2.4	8.04	7.2	98.5
3: highest fertility rates	289	0	0.4	3.57	1.3	5.76	4.3	50.1

4.5.2 For the 15 to 19 age group

4.5.2.1 2018-2019

Summary measures of each group for the 2018-2019 period and 15 to 19 age group can be seen in **Table 11**.

```r
# Creating the summary tables for the 15 to 19 age group for the 2018-2019 period
cria_tabelas(
  df = df_indicadores_18_19,
  var_grupos = "cluster_15_a_19",
  label_tabela = "3",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  nomes_legiveis = nomes_legiveis2,
  periodo = "2018 to 2019"
)
```

Table 11: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 15 to 19 (per thousand) in the period of 2018 to 2019.

Fertility rate (15 to 19 age group)

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2373	0.0	30.3	35.69	37.6	9.60	43.10	48.7
2: intermediary fertility rates	2442	48.8	54.5	61.87	60.9	8.65	68.60	80.2
3: highest fertility rates	755	80.4	85.8	98.78	92.8	19.66	105.55	269.7

HDI-M

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2373	0.48	0.66	0.70	0.71	0.06	0.74	0.86
2: intermediary fertility rates	2442	0.45	0.59	0.64	0.64	0.06	0.70	0.78
3: highest fertility rates	755	0.42	0.56	0.60	0.59	0.06	0.64	0.77

Primary Care coverage

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2373	0.0	89.70	90.97	99.8	16.66	100	100
2: intermediary fertility rates	2442	0.0	93.80	93.51	100.0	13.06	100	100
3: highest fertility rates	755	19.8	93.85	93.19	100.0	13.62	100	100

Percentage of women who are exclusive users of the SUS

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2373	0	3.0	13.42	8.7	12.99	20.6	84.3
2: intermediary fertility rates	2442	0	1.2	6.80	3.2	8.60	9.2	98.5
3: highest fertility rates	755	0	0.4	2.62	1.1	4.63	2.8	61.6

4.5.2.2 2020-2021

Summary measures of each group for the 2020-2021 period and 15 to 19 age group can be seen in **Table 12**.

```
# Creating the summary tables for the 15 to 19 age group for the 2020-2021 period
cria_tabelas(
  df = df_indicadores_20_21,
  var_grupos = "cluster_15_a_19",
  label_tabela = "4",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  nomes_legiveis = nomes_legiveis2,
  periodo = "2020 to 2021"
)
```

Table 12: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 15 to 19 (per thousand) in the period of 2020 to 2021.

Fertility rate (15 to 19 age group)

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2476	0.0	26.67	31.99	33.5	9.24	39.40	44.5
2: intermediary fertility rates	2423	44.6	49.80	57.15	55.8	8.57	63.90	75.4
3: highest fertility rates	671	75.5	80.75	93.77	87.2	19.57	100.55	230.2

HDI-M

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2476	0.49	0.66	0.70	0.71	0.06	0.74	0.86
2: intermediary fertility rates	2423	0.48	0.59	0.64	0.64	0.06	0.69	0.78
3: highest fertility rates	671	0.42	0.55	0.59	0.59	0.06	0.63	0.74

Primary Care coverage

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2476	21.6	87.5	90.81	99.6	15.52	100	100
2: intermediary fertility rates	2423	30.9	96.5	95.39	100.0	10.19	100	100
3: highest fertility rates	671	35.8	93.3	94.16	100.0	11.31	100	100

Percentage of women who are exclusive users of the SUS

Group (cluster)	n	Min.	1st Quartile	Mean	Median	S.D.	3rd Quartile	Max.
1: lowest fertility rates	2476	0	3.6	14.31	10.3	12.95	22.0	74.5
2: intermediary fertility rates	2423	0	1.2	6.66	3.3	8.22	9.3	98.5
3: highest fertility rates	671	0	0.4	2.46	1.0	3.97	2.6	30.9

4.6 Multiple comparisions tables

The code below contains a function that creates tables with the pairwise comparisions for the groups obtained for each variable considered.

```
# Creating a function that creates a table with the multiple comparisions
# Creating a function that creates a table with the multiple comparisions
cria_tabelas_testes <- function(
  variaveis,
  nomes_legiveis,
  df,
  var_grupos,
  label_tabela,
  faixa_etaria,
  periodo
) {
  for (variavel in variaveis) {
    formula <- as.formula(glue::glue("{variavel} ~ {var_grupos}"))
    resultados_dunn <- dunn_test(
      data = df,
      formula = formula,
      p.adjust.method = "bonferroni"
    )

    # Calculating Cohen's D
    efeito_d <- cohens_d(data = df, formula = formula)

    monta_linhas <- function(num_grupo) {
      conteudo <- paste(
        -1 *
          round(
            resultados_dunn$statistic[which(
              resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
            )],
            3
          ),
        "(Z) \\\\",
        ifelse(
          round(
            resultados_dunn$p.adj[which(
              resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
            )],
            3
          ) < 0.05,
          ifelse(
            round(
              resultados_dunn$p.adj[which(
                resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
              )],
              3
            ) == 0,
            "< 0.001*",
            paste0(
              round(
                resultados_dunn$p.adj[which(
                  resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
                )],
                3
              ),
```

```
          "*"
        )
      ),
      round(
        resultados_dunn$p.adj[which(
          resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
        )],
        3
      )
    ),
    "(p-value) \\\\",
    round(
      efeito_d$effsize[
        efeito_d$group1 == unique(efeito_d$group1)[num_grupo]
      ],
      3
    ),
    "(Cohen's D)"
  )

  glue::glue("\\makecell{{{conteudo}}}")
}

df_teste <- data.frame(
  grupo1 = monta_linhas(1),
  grupo2 = c(rep("", 1), monta_linhas(2)),
  row.names = unique(resultados_dunn$group2)
)

rownames(df_teste) <- paste0("\\textbf{", rownames(df_teste), "}")
colnames(df_teste) <- unique(resultados_dunn$group1)

if (variavel == variaveis[1]) {
  print(
    kbl(
      df_teste,
      align = "cc",
      caption = glue::glue(
        "Results of Dunn's tests (with Bonferroni correction) for multiple
        comparisons between pairs of groups of municipalities grouped by the
        fertility rate of women {faixa_etaria} (per thousand) in the period
        {periodo}. \\\\ \\\\ \\textbf{{{nomes_legiveis[[variavel]]}}}"
      ),
      label = paste0("tabela-comparacoes-", label_tabela),
      escape = FALSE
    ) |>
      kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
      row_spec(1, bold = TRUE)
  )
} else {
  cat(knitr::asis_output(paste0("**", nomes_legiveis[[variavel]], "**\n\n")))
  print(
    kbl(
      df_teste,
      align = "cc",
      escape = FALSE
    ) |>
      kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
```

```
        row_spec(0, bold = TRUE)
      )
    }
  }
}
nomes_variaveis1 <- c(
  "tx_fecundidade_10_a_14",
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus"
)
nomes_legiveis1 <- c(
  "Fertility rate (10 to 14 age group)",
  "HDI-M",
  "Primary Care coverage",
  "Percentage of women who are exclusive users of the SUS"
)
names(nomes_legiveis1) <- nomes_variaveis1


nomes_variaveis2 <- c(
  "tx_fecundidade_15_a_19",
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus"
)
nomes_legiveis2 <- c(
  "Fertility rate (15 to 19 age group)",
  "HDI-M",
  "Primary Care coverage",
  "Percentage of women who are exclusive users of the SUS"
)
names(nomes_legiveis2) <- nomes_variaveis2
```

4.6.1 For the 10 to 14 age group

4.6.1.1 2018-2019

Pairwise comparisions for the groups obtained for the 2018-2019 period and 10 to 14 age group can be seen in **Table 13**.

```
# Creating the multiple comparisions tables for the 10 to 14 age group and 2018-2019 period
cria_tabelas_testes(
  df = df_indicadores_18_19,
  var_grupos = "cluster_10_a_14",
  label_tabela = "1",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  nomes_legiveis = nomes_legiveis1,
  periodo = "2018 to 2019"
)
```

Table 13: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 10 to 14 (per thousand) in the period 2018 to 2019.

Fertility rate (10 to 14 age group)

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	-57.998 (Z) < 0.001* (p-value) -3.072 (Cohen's D)	
3: highest fertility rates	-44.066 (Z) < 0.001* (p-value) -2.943 (Cohen's D)	-14.252 (Z) < 0.001* (p-value) -1.855 (Cohen's D)

HDI-M

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	25.082 (Z) < 0.001* (p-value) 0.761 (Cohen's D)	
3: highest fertility rates	19.204 (Z) < 0.001* (p-value) 1.176 (Cohen's D)	6.305 (Z) < 0.001* (p-value) 0.437 (Cohen's D)

Primary Care coverage

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	-3.842 (Z) < 0.001* (p-value) -0.133 (Cohen's D)	
3: highest fertility rates	-2.645 (Z) 0.025* (p-value) -0.165 (Cohen's D)	-0.675 (Z) 1 (p-value) -0.031 (Cohen's D)

Percentage of women who are exclusive users of the SUS

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	20.469 (Z) < 0.001* (p-value) 0.546 (Cohen's D)	
3: highest fertility rates	18.798 (Z) < 0.001* (p-value) 0.973 (Cohen's D)	8.211 (Z) < 0.001* (p-value) 0.468 (Cohen's D)

4.6.1.2 2020-2021

Pairwise comparisions for the groups obtained for the 2020-2021 period and 10 to 14 age group can be seen in **Table 14**.

```
# Creating the multiple comparisions tables for the 10 to 14 age group and 2020-2021 period
cria_tabelas_testes(
  df = df_indicadores_20_21,
```

```
  var_grupos = "cluster_10_a_14",
  label_tabela = "2",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  nomes_legiveis = nomes_legiveis1,
  periodo = "2020 to 2021"
)
```

Table 14: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 10 to 14 (per thousand) in the period 2020 to 2021.

Fertility rate (10 to 14 age group)

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-58.468 (Z) < 0.001* (p-value) -3.075 (Cohen's D)	
3: highest fertility rates	-38.755 (Z) < 0.001* (p-value) -3.291 (Cohen's D)	-11.458 (Z) < 0.001* (p-value) -2.127 (Cohen's D)

HDI-M

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	27.561 (Z) < 0.001* (p-value) 0.854 (Cohen's D)	
3: highest fertility rates	17.199 (Z) < 0.001* (p-value) 1.168 (Cohen's D)	4.354 (Z) < 0.001* (p-value) 0.371 (Cohen's D)

Primary Care coverage

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-7.763 (Z) < 0.001* (p-value) -0.28 (Cohen's D)	
3: highest fertility rates	-2.77 (Z) 0.017* (p-value) -0.247 (Cohen's D)	0.796 (Z) 1 (p-value) 0.039 (Cohen's D)

Percentage of women who are exclusive users of the SUS

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	23.723 (Z) < 0.001* (p-value) 0.643 (Cohen's D)	
3: highest fertility rates	16.342 (Z) < 0.001* (p-value) 0.922 (Cohen's D)	5.251 (Z) < 0.001* (p-value) 0.31 (Cohen's D)

4.6.1.3 For the 15 to 19 age group

4.6.1.4 2018-2019

Pairwise comparisions for the groups obtained for the 2018-2019 period and 15 to 19 age group can be seen in **Table 15**.

```
# Creating the multiple comparisions tables for the 15 to 19 age group and 2018-2019 period
cria_tabelas_testes(
  df = df_indicadores_18_19,
  var_grupos = "cluster_15_a_19",
  label_tabela = "3",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  nomes_legiveis = nomes_legiveis2,
  periodo = "2018 to 2019"
)
```

Table 15: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 15 to 19 (per thousand) in the period 2018 to 2019.

Fertility rate (15 to 19 age group)

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	-51.938 (Z) < 0.001* (p-value) -2.865 (Cohen's D)	
3: highest fertility rates	-59.621 (Z) < 0.001* (p-value) -4.078 (Cohen's D)	-23.872 (Z) < 0.001* (p-value) -2.43 (Cohen's D)

HDI-M

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	25.962 (Z) < 0.001* (p-value) 0.831 (Cohen's D)	
3: highest fertility rates	32.317 (Z) < 0.001* (p-value) 1.616 (Cohen's D)	14.446 (Z) < 0.001* (p-value) 0.755 (Cohen's D)

Primary Care coverage

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	-3.738 (Z) 0.001* (p-value) -0.17 (Cohen's D)	
3: highest fertility rates	-3.505 (Z) 0.001* (p-value) -0.146 (Cohen's D)	-0.93 (Z) 1 (p-value) 0.024 (Cohen's D)

Percentage of women who are exclusive users of the SUS

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	19.936 (Z) < 0.001* (p-value) 0.601 (Cohen's D)	
3: highest fertility rates	29.538 (Z) < 0.001* (p-value) 1.108 (Cohen's D)	15.839 (Z) < 0.001* (p-value) 0.605 (Cohen's D)

4.6.1.5 2020-2021

Pairwise comparisions for the groups obtained for the 2020-2021 period and 15 to 19 age group can be seen in **Table 16**.

```r
# Creating the multiple comparisions tables for the 15 to 19 age group and 2020-2021 period
cria_tabelas_testes(
  df = df_indicadores_20_21,
  var_grupos = "cluster_15_a_19",
  label_tabela = "4",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  nomes_legiveis = nomes_legiveis2,
  periodo = "2020 to 2021"
)
```

Table 16: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 15 to 19 (per thousand) in the period 2020 to 2021.

Fertility rate (15 to 19 age group)

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-53.306 (Z) < 0.001* (p-value) -2.823 (Cohen's D)	
3: highest fertility rates	-57.104 (Z) < 0.001* (p-value) -4.038 (Cohen's D)	-22.053 (Z) < 0.001* (p-value) -2.424 (Cohen's D)

HDI-M

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	29.067 (Z) < 0.001* (p-value) 0.941 (Cohen's D)	
3: highest fertility rates	33.154 (Z) < 0.001* (p-value) 1.777 (Cohen's D)	14.028 (Z) < 0.001* (p-value) 0.799 (Cohen's D)

Primary Care coverage

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-10.393 (Z) < 0.001* (p-value) -0.349 (Cohen's D)	
3: highest fertility rates	-4.05 (Z) < 0.001* (p-value) -0.246 (Cohen's D)	2.767 (Z) 0.017* (p-value) 0.115 (Cohen's D)

Percentage of women who are exclusive users of the SUS

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	23.384 (Z) < 0.001* (p-value) 0.706 (Cohen's D)	
3: highest fertility rates	30.76 (Z) < 0.001* (p-value) 1.238 (Cohen's D)	15.37 (Z) < 0.001* (p-value) 0.65 (Cohen's D)

5 Phase 3: GAMLSS modeling

This stage of the article aimed to find and interpret a regression model that would best explain the fertility rate in women under 20 years of age in Brazilian municipalities, initially based on covariates such as the reference year, the HDI-M, the population coverage of Primary Care, the percentage of women aged 10 to 49 who are exclusive users of the SUS, and a variable indicating the Covid-19 pandemic. Although the interest was, most likely, in modeling the entire period from 2012 to 2021, we had to make the decision to restrict the analysis period to the years 2018 to 2021, since models that considered the entire period did not present appropriate adjustments. With this consideration in mind, a brief description of the variables that will be considered in this phase can be seen in **Table 17**.

Table 17: Description of the variables considered for the GAMLSS modeling.

Variable	Description
ano	Year
tx_fecundidade_menor_20	Fertility rate in women under 20 years of age
idhm	HDI-M
cobertura_ab	Population coverage of Primary Care
porc_exclusivas_sus	Percentage of women aged 10 to 49 who are exclusive users of the SUS
pandemia	Variable indicating the Covid-19 pandemic (receives 'yes' if the reference year is 2020 or 2021 and 'no' otherwise)

5.1 Preparing the data

```
# Reading and manipulating the data
df_indicadores_corrigido <- df_indicadores_corrigido |>
  filter(ano >= 2018 & ano <= 2021) |>
  mutate(
    codmunres = as.character(codmunres),
    tx_fecundidade_menor_20 = round(
      nvm_10_a_19_corrigido / pop_feminina_10_a_19 * 1000,
      1
```

```
  ),
  cobertura_ab = round(media_cobertura_ab / populacao_total * 100, 1),
  porc_exclusivas_sus = round(
    (pop_feminina_10_a_49 - pop_fem_10_49_com_plano_saude) /
      pop_feminina_10_a_49 *
      100,
    1
  ),
  pandemia = factor(
    ifelse(ano < 2020, "before", "during"),
    levels = c("before", "during")
  ),
  ano = as.factor(ano)
) |>
drop_na() # There are 5 municipalities without the HDI-M information
```

5.2 About regression

Initial attempts to find regression models always involve, of course, adjustments to linear regression models, within which we make assumptions that imply the normality of the response variable in question. The histogram of the fertility rate in women under 20 years old, presented in **Figure 27**, shows, however, a relatively common scenario in which the response variable of the problem **does not** appear to be normally distributed, which can be noted by the strong asymmetry to the right present in the graph. In this scenario, linear regression models have their assumptions violated and, therefore, their conclusions are no longer valid.

```
hist(
  df_indicadores_corrigido$tx_fecundidade_menor_20[which(df_indicadores_corrigido$ano == "2019")],
  freq = FALSE,
  col = "lightblue",
  main = "",
  xlab = "Fertility rate in women under 20 years old",
  ylab = "Density"
)
```

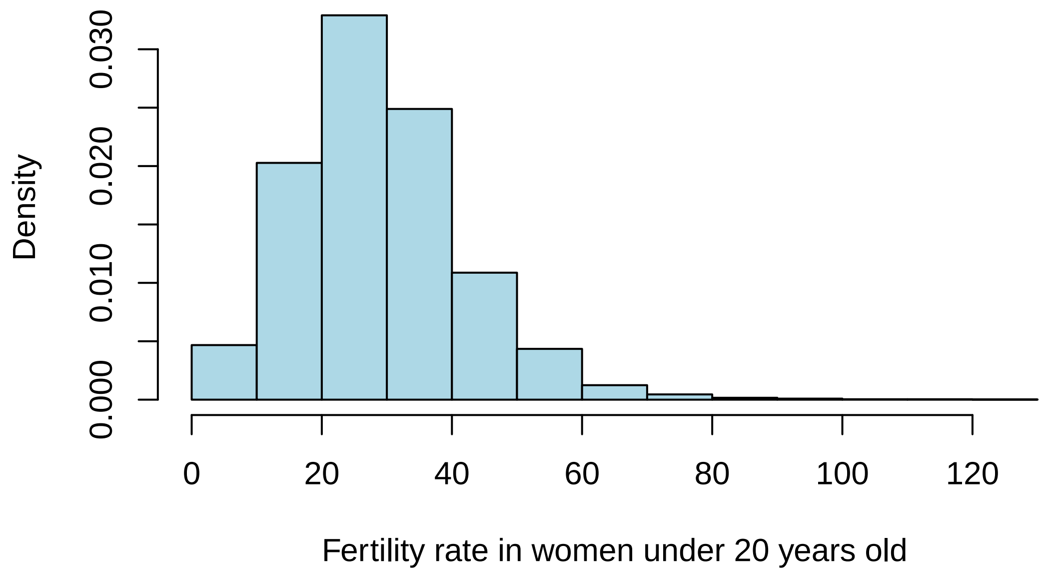

Figure 27: Histogram of the fertility rate in women under 20 years of age in 2019.

Thus, to overcome this problem, we expanded our search for regression models to GAMLSS models (R. A. Rigby and D. M. Stasinopoulos 2005), which allow us to choose from a range of different distributions for the response variable and which have as a major differential the fact that they allow us to separately model the parameters of location (related to the mean of the distribution), scale (related to the variance of the distribution) and shape (related to the asymmetry and kurtosis of the distribution). We adjusted and verified the adequacy of a series of different GAMLSS models, and decided to discuss in this work the models that: 1) assume that the response variable follows a skew t type 4 distribution, which allows us to model all four parameters discussed; and 2) assume that the response variable follows a zero-adjusted gamma distribution (ZAGA), which allows us to model the parameters of location, scale and the shape parameter related to the asymmetry of the distribution. It is important to note that, to avoid multicollinearity problems, we decided not to include in the analysis the covariate of the percentage of women aged 10 to 49 who are exclusive users of the SUS, since, as can be seen in the correlogram shown in **Figure 28**, the variables `porc_exclusivas_sus` and `idhm` have a considerable linear correlation.

```
corrplot(
  cor(
    df_indicadores_corrigido |>
      filter(ano == "2019") |>
      dplyr::select_at(vars(
        starts_with("tx") | "idhm" | "cobertura_ab",
        "porc_exclusivas_sus"
      )),
    use = "complete.obs",
    method = "pearson"
  ),
  method = "color",
  type = "lower",
  addCoef.col = "black",
```

```
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 0.85,
  diag = TRUE,
  cl.pos = "b",
  mar = c(0, 0, 1.5, 0)
)
```

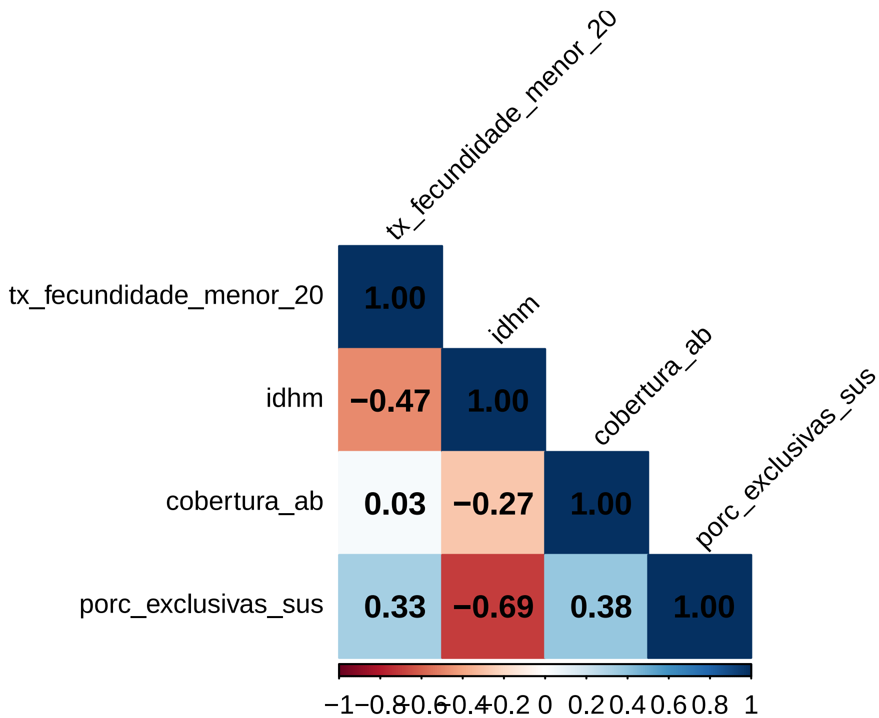

Figure 28: Pearson correlations between the variables in the dataset in 2019.

5.3 Fitting the models

Since GAMLSS models give us the freedom to model separately the location, scale and shape parameters of the distribution, a very large set of different models for a chosen distribution can be fitted. Therefore, after assuming a distribution for the response variable, we need to find the best model among all those that could be fitted considering this chosen distribution. To do this, we used two strategies described by Stasinopoulos et al. (2017, 397), which, in short, use stepwise selection methods and 1) allow different sets of covariates to be chosen to model each parameter of the distribution; or 2) force the same covariates to be selected to model all parameters of the distribution. Therefore, for each of the two selected distributions (skew t type 4 and ZAGA), we fitted:

- the full model, which has a random intercept for the reference year and which uses as covariates `coverage_ab`, `idhm`, `pandemic` and their interactions;
- the reduced model chosen by strategy 1;
- the reduced model chosen by strategy 2;
- and a third reduced model, which uses, for the μ model, only the significant terms of the full model.

To decide whether the reduced models were better than the full model, we used the generalized likelihood ratio test (GLRT) - which, in a simplified way, tests the null hypothesis that the reduced model is better than the full model (against the hypothesis that the full model is better). The **rejection** of the GLR null hypothesis, then, gives us evidence to believe that the reduced model is worse than the full model (which would therefore lead us to choose the full model). If the GLRT null hypothesis was not rejected for all the reduced models, we chose the reduced model that presented the most appropriate residual analysis or the lowest AIC value. Finally, after selecting the best model for each distribution, we chose, between the two selected models, the one that presented the most appropriate residual analysis or the lowest AIC value. All analyses were conducted using functions from the `{gamlss}` package (R. A. Rigby and D. M. Stasinopoulos 2005).

It is important to note that we chose to incorporate a random effect for the reference year to address the issue that observations from the same year are possibly correlated with each other, which violates the basic assumption that the observations are all independent.

5.3.1 For the skew t type 4 distribution

```
# Adjusting the full model
fit1_st4 <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  data = df_indicadores_corrigido,
  family = ST4(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
saveRDS(fit1_st4, "r_objects/02_under_registration_corrected/fit1_st4.RDS")

sumario_fit1_st4 <- summary(fit1_st4, save = TRUE)
saveRDS(sumario_fit1_st4, "r_objects/02_under_registration_corrected/sumario_fit1_st4.RDS")

# Utilizing the two strategies described in the book "Flexible Regression and
# Smoothing: Using GAMLSS in R", page 397, to get the best models for each parameter
## Adjusting a model with only the intercept for mu and considering sigma and nu
## as constants
fit2_st4 <- gamlss(
  tx_fecundidade_menor_20 ~ 1,
  data = df_indicadores_corrigido,
  family = ST4,
  control = gamlss.control(n.cyc = 500)
)

## Strategy 1:
### Starting from the simplest model and utilizing the forward stepwise method
### to find the model with the lowest AIC for each parameter
### Obs.: in this strategy, different variables can be selected for each
### parameter's model
fit3_st4 <- stepGAICAll.A(
  fit2_st4,
  scope = list(
```

```r
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
saveRDS(fit3_st4, "r_objects/02_under_registration_corrected/fit3_st4.RDS")

sumario_fit3_st4 <- summary(fit3_st4, save = TRUE)
saveRDS(sumario_fit3_st4, "r_objects/02_under_registration_corrected/sumario_fit3_st4.RDS")

### Strategy 2:
### This strategy forces all os the models to select the same variables subsets
fit4_st4 <- stepGAICAll.B(
  fit2_st4,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
saveRDS(fit4_st4, "r_objects/02_under_registration_corrected/fit4_st4.RDS")

sumario_fit4_st4 <- summary(fit4_st4, save = TRUE)
saveRDS(sumario_fit4_st4, "r_objects/02_under_registration_corrected/sumario_fit4_st4.RDS")

## Adjusting a third model with only the significant terms for mu
fit5_st4 <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      pandemia +
      cobertura_ab +
      idhm +
      pandemia * cobertura_ab +
      cobertura_ab * idhm,
  data = df_indicadores_corrigido,
  family = ST4(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
saveRDS(fit5_st4, "r_objects/02_under_registration_corrected/fit5_st4.RDS")

sumario_fit5_st4 <- summary(fit5_st4, save = TRUE)
saveRDS(sumario_fit5_st4, "r_objects/02_under_registration_corrected/sumario_fit5_st4.RDS")
```

Tables 18 to **21** show the estimates, standard errors, and significance tests for the models adjusted for μ in each of the four settings, as well as the additional random intercept terms for each year in each model. The applications of the GLRT and a comparison between the AIC of the three models can be seen below the tables.

- **For the complete model:**

Table 18: Measures related to the estimated coefficients for the location parameter in the full model (for the skew t type 4 distribution).

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	140.5131	4.2845	32.7955	0.0000
pandemiaduring	-4.7870	1.9133	-2.5020	0.0124
cobertura_ab	-0.6043	0.0451	-13.3979	0.0000
idhm	-153.0933	5.6863	-26.9230	0.0000
pandemiaduring:cobertura_ab	0.0245	0.0075	3.2589	0.0011
pandemiaduring:idhm	-0.4668	2.1531	-0.2168	0.8284
cobertura_ab:idhm	0.7458	0.0608	12.2574	0.0000

```
##      2018       2019       2020       2021
##  0.8303717 -0.8311686  0.1901320 -0.1915618
```

- **For the reduced model chosen by the first variable selection strategy:**

Table 19: Measures related to the estimated coefficients for the location parameter in the first reduced model (for the skew t type 4 distribution).

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	137.8508	4.1772	33.0009	0
idhm	-152.8935	5.5510	-27.5436	0
cobertura_ab	-0.5885	0.0442	-13.3024	0
idhm:cobertura_ab	0.7403	0.0594	12.4611	0

```
##      2018       2019       2020       2021
##  2.3839170  0.6359364 -1.3198620 -1.6301453
```

- **For the reduced model chosen by the second variable selection strategy:**

Table 20: Measures related to the estimated coefficients for the location parameter in the second reduced model (for the skew t type 4 distribution).

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	139.4981	4.2618	32.7326	0
idhm	-155.3825	5.6499	-27.5020	0
cobertura_ab	-0.6062	0.0453	-13.3974	0
idhm:cobertura_ab	0.7670	0.0606	12.6550	0

```
##      2018       2019       2020       2021
##  2.3725630  0.6248624 -1.2643131 -1.6689844
```

- **For the reduced model chosen by the third variable selection strategy:**

Table 21: Measures related to the estimated coefficients for the location parameter in the third reduced model (for the skew t type 4 distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	140.6442	4.1807	33.6412	0.0000
pandemiaduring	-5.1840	0.5536	-9.3645	0.0000
cobertura_ab	-0.6039	0.0444	-13.5909	0.0000
idhm	-153.2347	5.5679	-27.5209	0.0000
pandemiaduring:cobertura_ab	0.0254	0.0064	3.9699	0.0001
cobertura_ab:idhm	0.7447	0.0598	12.4578	0.0000

```
##       2018       2019       2020       2021
##  2.3725630  0.6248624 -1.2643131 -1.6689844
```

- **GLRT applications and comparison between AICs:**

```
# Using the LR test to verify if the simpler models are better than the full model
## The model selected by the first strategy IS better than the full model
LR.test(fit3_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168275.4 with  26.23313 deg. of freedom
##  Altenative model: deviance= 168258.2 with  37.99771 deg. of freedom
##
##  LRT = 17.22161 with 11.76458 deg. of freedom and p-value= 0.1313122
```

```
## The model selected by the second strategy IS better than the full model
LR.test(fit4_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168285.7 with  29.12117 deg. of freedom
##  Altenative model: deviance= 168258.2 with  37.99771 deg. of freedom
##
##  LRT = 27.48906 with 8.876539 deg. of freedom and p-value= 0.001069358
```

```
## The model selected by the third strategy IS better than the full model
LR.test(fit5_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168258.2 with  36.99923 deg. of freedom
##  Altenative model: deviance= 168258.2 with  37.99771 deg. of freedom
##
##  LRT = 0.05195196 with 0.9984853 deg. of freedom and p-value= 0.8191952
```

```r
# Comparing the AICs
GAIC(fit1_st4, fit3_st4, fit4_st4, fit5_st4)
```

```
##                df      AIC
## fit3_st4 26.23313 168327.9
## fit5_st4 36.99923 168332.2
## fit1_st4 37.99771 168334.2
## fit4_st4 29.12117 168343.9
```

As we can see, the application of GLRT to compare the full model with the reduced models gives us evidence to believe that the reduced models chosen by the first and third strategy are better than the full model. As for the decision of which reduced model to choose, we can base ourselves on both the AIC values and a new application of GLRT, which can be seen below (this application can be done since the models chosen by the first and second strategies are nested in the one chosen by the third). In either case, we would choose the reduced model chosen by the first strategy, `fit3_st4`.

```r
# Since the fit3_st4 model is nested within fit5_st4, we can compare them using
# GLRT
LR.test(fit3_st4, fit5_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168275.4 with  26.23313 deg. of freedom
##  Altenative model: deviance= 168258.2 with  36.99923 deg. of freedom
##
##  LRT = 17.16966 with 10.7661 deg. of freedom and p-value= 0.09469045
```

However, it would be in our interest to maintain the interaction term between the Covid-19 pandemic indicator variable and Primary Care coverage, since it provides us with useful interpretations. Thus, despite presenting a slightly larger AIC (and having the null hypothesis of GLRT with fit3_st4 not rejected), we opted for the model chosen by the third selection strategy, `fit5_st4`.

Typical graphs related to the analysis of residuals of this model can be seen in **Figure 29**. It is important to note here that the residuals considered by the GAMLSS models are the so-called *normalized (randomized) quantile residuals*, which, **when the distribution assumed for the response variable is adequate**, always follow a standard normal distribution. As we can see by observing the graphs presented in this figure, the assumptions of the model seem to be respected, since:

1. in the first two graphs, of residuals against adjusted values and of residuals against the indexes of the observations, the residuals appear to be distributed randomly around the horizontal line at $y = 0$;
2. in the third graph, of the estimated kernel density of the residuals, the density resembles the density of the standard normal distribution;
3. in the fourth graph, of the normal Q-Q Plot of the residuals, the observed quantiles of the residuals are in agreement with the theoretical quantiles of the standard normal distribution (to a large extent).

Furthermore, observing the summary measures present in the R output below the graphs, we can see that the mean of the residuals is approximately zero, their variance is approximately one, their skewness coefficient is close to zero and their kurtosis coefficient is close to 3. Thus, both the graphs and the summary statistics suggest that the residuals follow, approximately, the standard normal distribution, which would be necessary for the model to be considered adequate.

```r
# Residual analysis of the selected model
plot(fit5_st4)
```

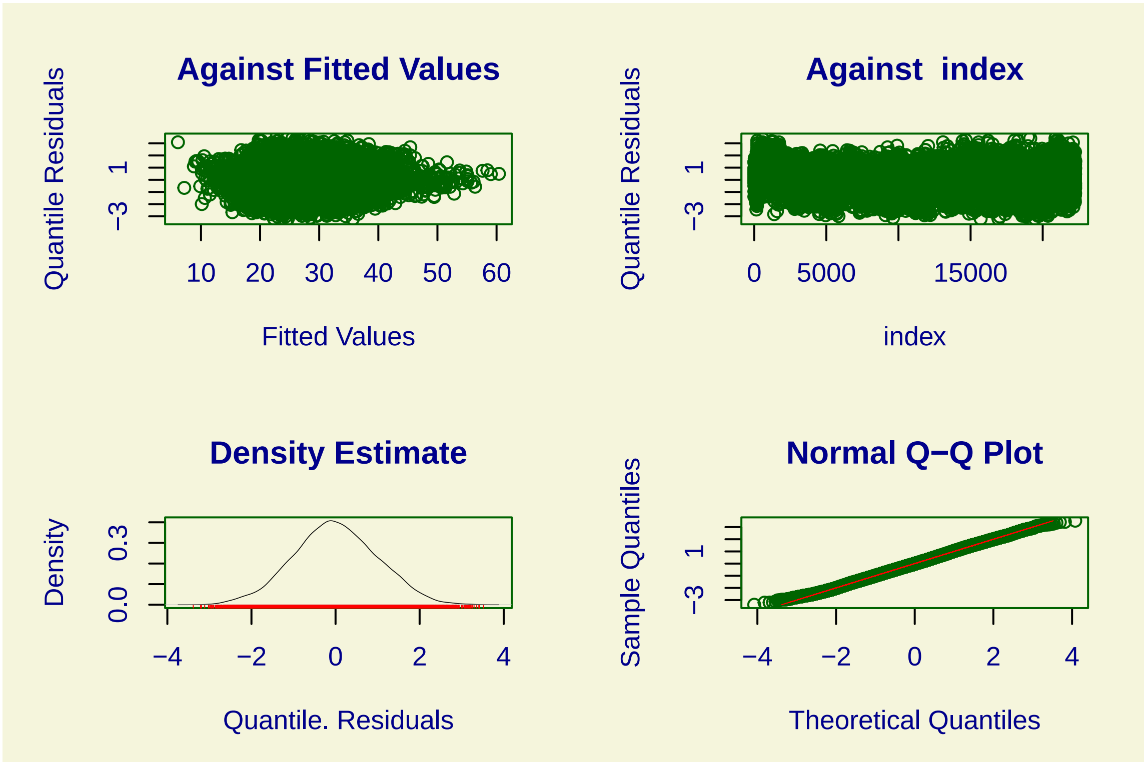

Figure 29: Usual graphs related to the residual analysis of the selected model for the skew t type 4 distribution.

```
## ******************************************************************
##	     Summary of the Quantile Residuals
##                            mean   =  0.001743274
##                        variance   =  0.9981581
##                coef. of skewness  =  0.03521314
##                coef. of kurtosis  =  2.873077
## Filliben correlation coefficient  =  0.9997277
## ******************************************************************
```

An additional check of the model's adequacy can be done through *worm plots*. In simplified terms, we would expect an adequate model that:

- the yellow points on the graph, which represent the model's residuals, are as close as possible to the dotted horizontal line at $y = 0$, which represents their approximate expected values;
- 95% of the yellow points are between the elliptical curves on the graph, since they represent an approximate 95% confidence interval (*point-wise*) for the model's residuals;
- and that the curve fitted to the points on the graph (in red) is as straight as possible and is as close as possible to the horizontal line at $y = 0$.

As we can see, the graph shown in **Figure 30** does indeed present points that are outside the graph's elliptical curves, which reveal that the fit was not as perfect as one would like it to be. However, as the deviations do not appear to be so numerous, and as other distributions resulted in adjustments that were extremely inferior to what was obtained through the asymmetric t distribution of type 4, we chose to consider the model found as adequate (as far as possible).

```r
wp(fit5_st4, ylim.all = 0.6)
```

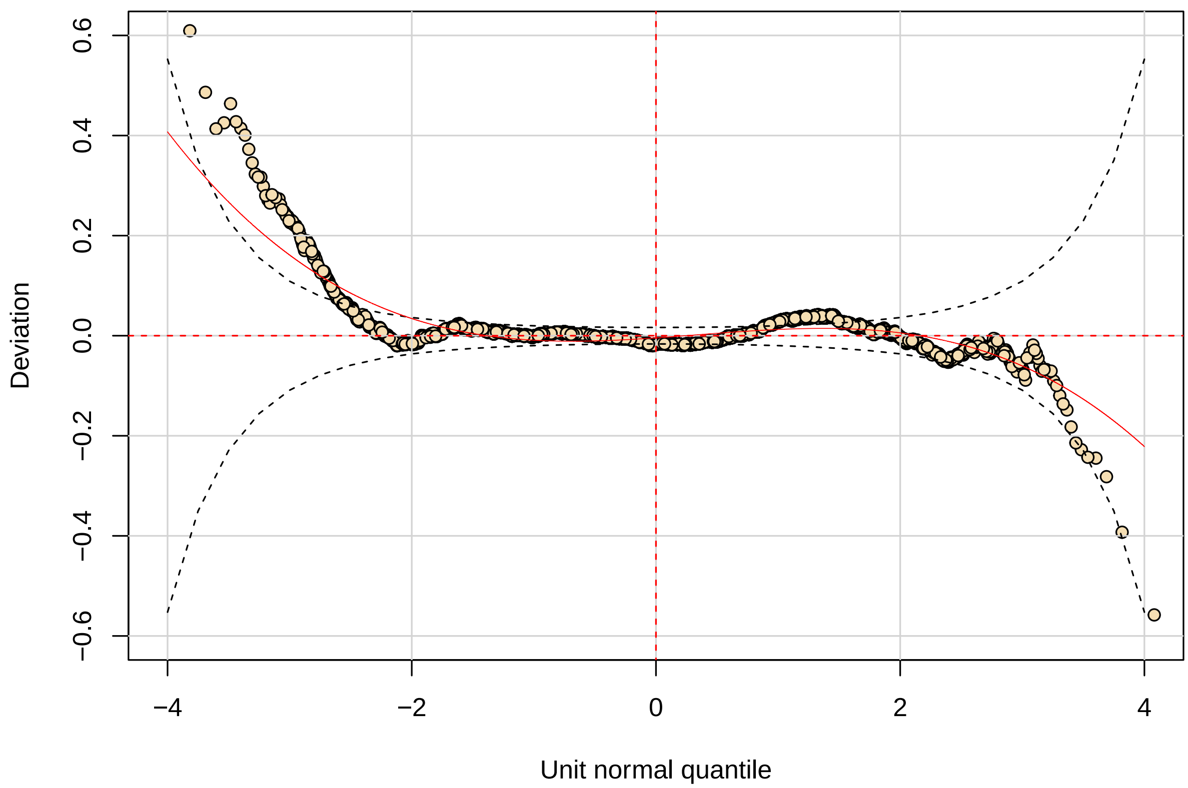

Figure 30: Worm plot of the selected model for the skew t type 4 distribution.

5.3.2 For the ZAGA distribution

```r
# Adjusting the full model
fit1_zaga <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  data = df_indicadores_corrigido,
  family = ZAGA(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
```

```
)
saveRDS(fit1_zaga, "r_objects/02_under_registration_corrected/fit1_zaga.RDS")

sumario_fit1_zaga <- summary(fit1_zaga, save = TRUE)
saveRDS(sumario_fit1_zaga, "r_objects/02_under_registration_corrected/sumario_fit1_zaga.RDS")

# Utilizing the two strategies described in the book "Flexible Regression and
# Smoothing: Using GAMLSS in R", page 397, to get the best models for each parameter
## Adjusting a model with only the intercept for mu and considering sigma and nu
## as constants
fit2_zaga <- gamlss(
  tx_fecundidade_menor_20 ~ 1,
  data = df_indicadores_corrigido,
  family = ZAGA,
  control = gamlss.control(n.cyc = 500)
)

## Strategy 1:
### Starting from the simplest model and utilizing the forward stepwise method to
### find the model with the lowest AIC for each parameter
### Obs.: in this strategy, different variables can be selected for each
### parameter's model
fit3_zaga <- stepGAICAll.A(
  fit2_zaga,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
saveRDS(fit3_zaga, "r_objects/02_under_registration_corrected/fit3_zaga.RDS")

sumario_fit3_zaga <- summary(fit3_zaga, save = TRUE)
saveRDS(sumario_fit3_zaga, "r_objects/02_under_registration_corrected/sumario_fit3_zaga.RDS")

## Strategy 2:
### This strategy forces all os the models to select the same variables subsets
fit4_zaga <- stepGAICAll.B(
  fit2_zaga,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
saveRDS(fit4_zaga, "r_objects/02_under_registration_corrected/fit4_zaga.RDS")

sumario_fit4_zaga <- summary(fit4_zaga, save = TRUE)
saveRDS(sumario_fit4_zaga, "r_objects/02_under_registration_corrected/sumario_fit4_zaga.RDS")

## Adjusting a third model with only the significant terms for mu
fit5_zaga <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      pandemia +
      cobertura_ab +
      idhm +
      pandemia * idhm +
```

```
      cobertura_ab * idhm,
  data = df_indicadores_corrigido,
  family = ZAGA(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
saveRDS(fit5_zaga, "r_objects/02_under_registration_corrected/fit5_zaga.RDS")

sumario_fit5_zaga <- summary(fit5_zaga, save = TRUE)
saveRDS(sumario_fit5_zaga, "r_objects/02_under_registration_corrected/sumario_fit5_zaga.RDS")
```

Tables 22 to **25** show the estimates, standard errors, and significance tests for the models adjusted for μ in each of the four settings, as well as the additional random intercept terms for each year in each model. The applications of the GLRT and a comparison between the AIC of the three models can be seen below the tables.

- **For the complete model:**

Table 22: Measures related to the estimated coefficients for the location parameter in the full model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.4741	0.1622	46.0793	0.0000
pandemiaduring	0.1353	0.0696	1.9432	0.0520
cobertura_ab	-0.0234	0.0017	-13.7814	0.0000
idhm	-5.7761	0.2232	-25.8822	0.0000
pandemiaduring:cobertura_ab	0.0003	0.0003	1.0427	0.2971
pandemiaduring:idhm	-0.4016	0.0791	-5.0777	0.0000
cobertura_ab:idhm	0.0310	0.0024	13.0956	0.0000

```
##         2018         2019         2020         2021
##  0.029027169 -0.029080204 -0.001315118  0.001300009
```

- **For the reduced model chosen by the first variable selection strategy:**

Table 23: Measures related to the estimated coefficients for the location parameter in the first reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.5204	0.1606	46.8127	0
idhm	-5.9429	0.2220	-26.7732	0
cobertura_ab	-0.0230	0.0017	-13.5404	0
idhm:cobertura_ab	0.0306	0.0024	12.9156	0

```
##        2018        2019        2020        2021
##  0.07902994  0.01828129 -0.05094175 -0.04690787
```

- **For the reduced model chosen by the second variable selection strategy:**

Table 24: Measures related to the estimated coefficients for the location parameter in the second reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.4588	0.1610	46.3359	0.0000
idhm	-5.7720	0.2228	-25.9089	0.0000
cobertura_ab	-0.0233	0.0017	-13.7776	0.0000
pandemiaduring	0.1871	0.0478	3.9123	0.0001
idhm:cobertura_ab	0.0311	0.0024	13.1574	0.0000
idhm:pandemiaduring	-0.4337	0.0722	-6.0096	0.0000

```
##         2018         2019         2020         2021
##  0.028982241 -0.029038166 -0.001653589  0.001635906
```

- **For the reduced model chosen by the third variable selection strategy:**

Table 25: Measures related to the estimated coefficients for the location parameter in the third reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.4561	0.1610	46.3223	0.0000
pandemiaduring	0.1878	0.0479	3.9250	0.0001
cobertura_ab	-0.0233	0.0017	-13.7533	0.0000
idhm	-5.7681	0.2227	-25.8969	0.0000
pandemiaduring:idhm	-0.4348	0.0722	-6.0207	0.0000
cobertura_ab:idhm	0.0310	0.0024	13.1337	0.0000

```
##         2018         2019         2020         2021
##  0.028982460 -0.029038382 -0.001680011  0.001664729
```

- **GLRT applications and comparison between AICs:**

```r
# Using the LR test to verify if the simpler models are better than the full model
## The model selected by the first strategy (fit3_zaga) ISN'T better than the full model
LR.test(fit3_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168466.1 with  19.68266 deg. of freedom
##  Altenative model: deviance= 168426.4 with  25.74267 deg. of freedom
##
##  LRT = 39.68176 with 6.060017 deg. of freedom and p-value= 0.0000005604524
```

```r
## The model selected by the second strategy (fit4_zaga) IS better than the full model
LR.test(fit4_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168429.4 with  22.77467 deg. of freedom
##  Altenative model: deviance= 168426.4 with  25.74267 deg. of freedom
##
##  LRT = 2.93669 with 2.968004 deg. of freedom and p-value= 0.3959278
```

```r
## The model selected by the third strategy (fit5_zaga) IS better than the full model
LR.test(fit5_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168427.5 with  24.74173 deg. of freedom
##  Altenative model: deviance= 168426.4 with  25.74267 deg. of freedom
##
##  LRT = 1.078339 with 1.000947 deg. of freedom and p-value= 0.2993646
```

```r
# Comparing the AICs
GAIC(fit1_zaga, fit3_zaga, fit4_zaga, fit5_zaga)
```

```
##                 df      AIC
## fit4_zaga 22.77467 168474.9
## fit5_zaga 24.74173 168477.0
## fit1_zaga 25.74267 168477.9
## fit3_zaga 19.68266 168505.5
```

As we can see, the application of GLRT to compare the full model with the reduced models gives us evidence to believe that the reduced models chosen by the second and third strategy are better than the full model. Again, as for the decision of which reduced model to choose, we can base ourselves on both the AIC values and a new application of GLRT, which can be seen below (this application can be done since the model chosen by the second strategy is nested in the one chosen by the third). In either case, we choose the reduced model chosen by the second strategy, `fit4_zaga`.

```r
# Since the fit4_zaga model is nested within fit5_zaga, we can compare them
# using GLRT (both have the same model for mu, but fit5_zaga has the complete
# model for all other parameters)
LR.test(fit4_zaga, fit5_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 168429.4 with  22.77467 deg. of freedom
##  Altenative model: deviance= 168427.5 with  24.74173 deg. of freedom
##
##  LRT = 1.858351 with 1.967057 deg. of freedom and p-value= 0.3875183
```

As in the case of the type 4 asymmetric t distribution, typical graphs related to the residual analysis of this model can be seen **Figure 31**. As we can see by observing the graphs presented in this figure and the summary measures present in the R output below them, the model's assumptions do not seem to be being respected. This statement becomes evident when we observe the normal Q-Q Plot of the residuals - which has many values far from the red line as we approach the right tail of the distribution, indicating that the distribution of the residuals appears to be skewed to the right - and when we observe its kurtosis coefficient, which is considerably greater than 3, indicating that the distribution of the residuals appears to have heavier tails than the standard normal distribution.

```r
# Residual analysis of the selected model
plot(fit4_zaga)
```

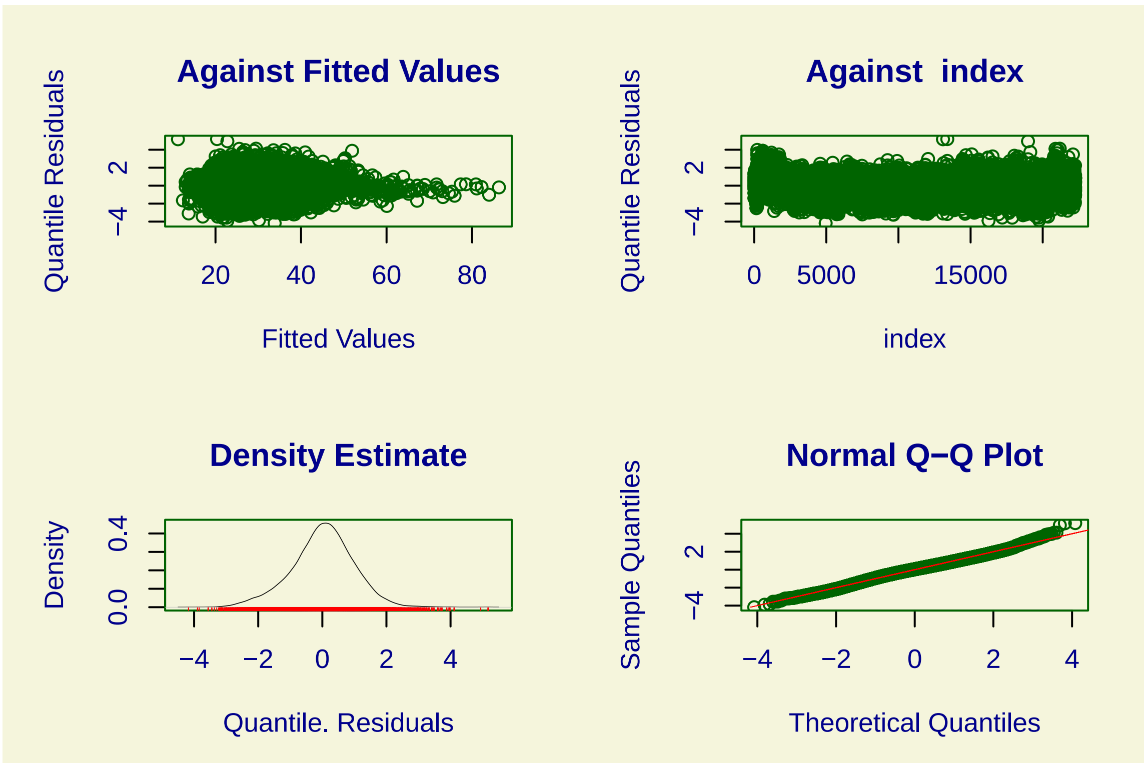

Figure 31: Usual graphs related to the residual analysis of the selected model for the ZAGA distribution.

```
## ******************************************************************
##   Summary of the Randomised Quantile Residuals
##                            mean   =  0.008267511
##                        variance   =  0.9576244
##                coef. of skewness  =  -0.1569567
##                coef. of kurtosis  =  3.524258
## Filliben correlation coefficient  =  0.9972839
## ******************************************************************
```

Furthermore, observing the worm plot shown in **Figure 32**, we can see that most of the points are outside the space formed between the two elliptic curves, giving us even more evidence that the model's residuals do not follow a standard normal distribution and that, therefore, the model is not suitable. Therefore, we conducted the remaining analyses using the reduced model chosen for the skew t type 4 distribution, since its residual analyses showed that that model appears to be, with its reservations, the most suitable.

```r
wp(fit4_zaga, ylim.all = 0.6)
```

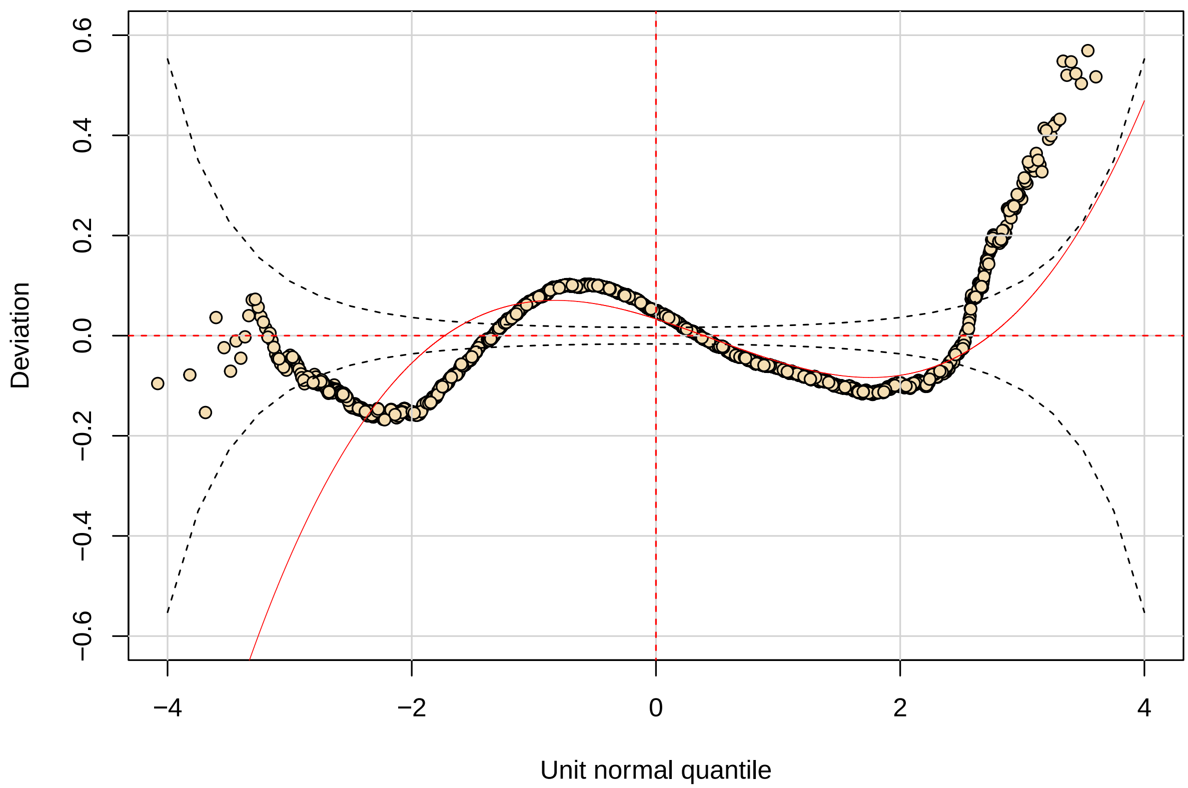

Figure 32: Worm plot of the selected model for the ZAGA distribution.

5.4 Interpreting the selected model

```r
fit_final <- fit5_st4
sumario_fit_final <- sumario_fit5_st4
```

Starting the analysis of the selected model, which assumes that the fertility rate in women under 20 years of age follows a skew t type 4 distribution, we present, in **Table 26**, the estimates, standard errors and hypothesis tests for the coefficients of the model adjusted for the location parameter - related to the mean of the distribution of the response variable - followed by the additional terms of random intercepts for each year of the model.

```r
kbl(
  round(sumario_fit_final$coef.table[c(1:6), ], 3),
  align = "cccc",
  caption = "Measures related to the estimated coefficients of the model for the
    location parameter."
) |>
```

```
kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 26: Measures related to the estimated coefficients of the model for the location parameter.

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	140.644	4.181	33.641	0
pandemiaduring	-5.184	0.554	-9.365	0
cobertura_ab	-0.604	0.044	-13.591	0
idhm	-153.235	5.568	-27.521	0
pandemiaduring:cobertura_ab	0.025	0.006	3.970	0
cobertura_ab:idhm	0.745	0.060	12.458	0

```
getSmo(fit_final)$coef
```

```
##       2018       2019       2020       2021
##  0.8306829 -0.8312702  0.1911875 -0.1921879
```

As we can see from the sign of the estimates (and from the results of the coefficient significance tests), we can say that:

- individually, the increase in the HDI-M generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- individually, the increase in Primary Care coverage generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- individually, being a year of the Covid-19 pandemic generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- for years of the COVID-19 pandemic, the increase in PCP coverage generates an increase in the value of the fertility rate in women under 20 years of age in the municipality;
- the joint increase in the HDI-M and the PCP coverage generates an increase in the value of the municipality's fertility rate.

However, unlike linear regression models, in which making more informative interpretations about the effect of each covariate on the response variable is simple and straightforward, similar interpretations from GAMLSS models are not available. Therefore, in order to better understand how the fertility rate in women under 20 years of age behaves according to the variation of the covariates in the data set, we used the following strategy:

- we set 3 HDI-M values (quantile 0.25, median value and quantile 0.75, resulting in HDI-M values of, respectively, 0.599, 0.665 and 0.718);
- for each HDI-M value set, we varied the value of the reference year (2018 to 2021) and, therefore, the value of the variable indicating the Covid-19 pandemic;
- for each value of the HDI-M, the year and the variable indicating the Covid-19 pandemic, we used the selected model to obtain predictions for the value of the fertility rate that a municipality with Primary Care coverage ranging from 0%, 10%, 20%, . . . , 100% would have;

- We constructed a line graph that receives the value of Primary Care coverage on the x axis and the predicted value of the fertility rate on the y axis, whose lines represent the predictions for each of the three HDI-M values set in the corresponding period (the "non-pandemic" period being an average of the years 2018 and 2019 and the "with pandemic" period being an average of the years 2020 and 2021).

The graphs resulting from this process can be seen in **Figure 33**. The average variation in the predicted values of the fertility rate caused by the increase of 10 units in the population coverage of AB for each value of the HDI-M and the pandemic indicator variable can be seen in **Table 27**.

```
# Trying to better understand the impact of each variable on the fertility rate
## Creating a dataframe with the desired observations to predict
idhms <- c(
  as.numeric(quantile(df_indicadores_corrigido$idhm, 0.25)),
  median(df_indicadores_corrigido$idhm),
  as.numeric(quantile(df_indicadores_corrigido$idhm, 0.75))
)
anos <- 2018:2021
coberturas_ab <- seq(0, 100, by = 10)

df_newdata <- expand.grid(idhms, anos, coberturas_ab) |>
  arrange(Var1) |>
  mutate(
    pandemia = factor(
      ifelse(Var2 < 2020, "before", "during"),
      levels = c("before", "during")
    )
  )

colnames(df_newdata) <- c("idhm", "ano", "cobertura_ab", "pandemia")

## Making the predictions
predicoes <- predict(fit_final, newdata = df_newdata, type = "response")
```

```
## New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```

```
df_newdata_completo <- df_newdata |>
  mutate(
    idhm = as.factor(idhm),
    predicoes = predicoes
  ) |>
  group_by(idhm, cobertura_ab, pandemia) |>
  summarise(predicoes = mean(predicoes))
```

```
## Plotting the predictions
ggplot(
  data = df_newdata_completo,
  mapping = aes(
    x = cobertura_ab,
    y = predicoes,
    colour = idhm,
    linetype = pandemia
  )
) +
  geom_line(linewidth = 1) +
  geom_point() +
  scale_x_continuous(breaks = unique(df_newdata_completo$cobertura_ab)) +
```

```r
  labs(
    x = "Primary care coverage",
    y = "Predicted fertility rate per thousand women aged 10 to 19",
    colour = "HDI-M",
    linetype = "COVID-19 pandemic"
  ) +
  theme_bw(base_size = 14) +
  theme(legend.position = "bottom")
```

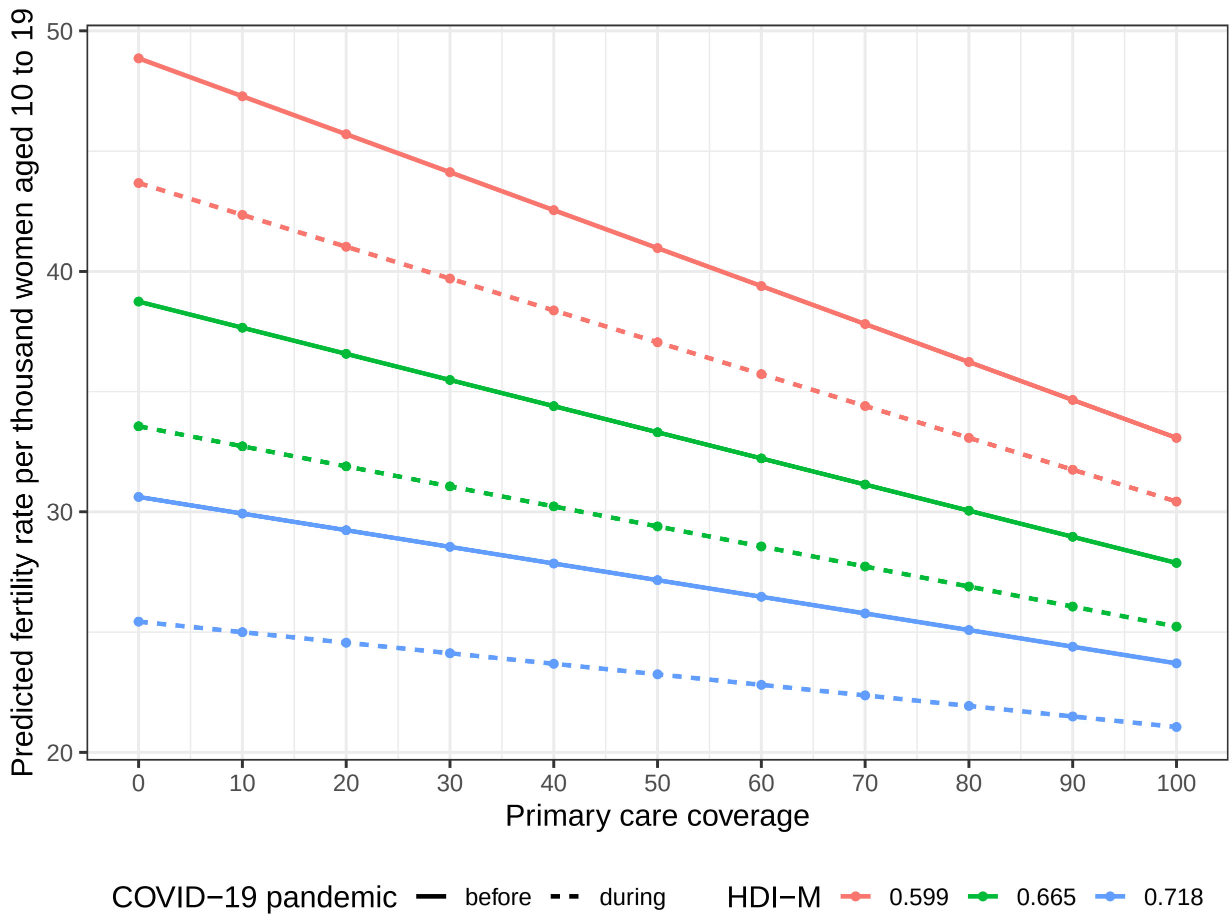

Figure 33: Predicted values of the fertility rate of women under 20 years of age according to HDI-M values, Primary Care Population coverage and information on the year of the pandemic. Brazil, 2018-2021.

```r
## Creating a dataframe with the mean variations of the predicted fertility rates
## for each combination of HDI-M and the pandemic indicator when the primary care
## coverage is increased by 10
df_variacao <- df_newdata_completo |>
  group_by(idhm, pandemia) |>
  mutate(variacao = (predicoes - lag(predicoes))) |>
  summarise(variacao_media = round(mean(variacao, na.rm = T), 3)) |>
  arrange(idhm, pandemia)

kbl(
  df_variacao,
  align = "ccc",
```

```r
  col.names = c(
    "HDI-M",
    "Covid-19 pandemic",
    "Mean variation on the predictions"
  ),
  caption = "Average variation in predicted fertility rate values caused by an
    increase of 10 units in Primary Care coverage for each fixed value of the
    HDI-M and the pandemic indicator variable."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 27: Average variation in predicted fertility rate values caused by an increase of 10 units in Primary Care coverage for each fixed value of the HDI-M and the pandemic indicator variable.

HDI-M	Covid-19 pandemic	Mean variation on the predictions
0.599	before	-1.578
0.599	during	-1.324
0.665	before	-1.087
0.665	during	-0.833
0.718	before	-0.692
0.718	during	-0.438

Analyzing the estimates used to construct the graph presented in more detail, we can say, based on the table above and descriptively, that:

- For a municipality with an HDI-M in the 0.25 quantile of the HDI-Ms of Brazilian municipalities (from 0.599), an increase of 10 units in population coverage with Family Health teams generates:
    1. for years not affected by the Covid-19 pandemic, an average decrease of 1.578 units in the fertility rate of women under 20 years of age in the municipality;
    2. for years affected by the pandemic, an average decrease of 1.324 units in this same rate.
- For a municipality with the median HDI-M among municipalities in Brazil (of 0.665), an increase of 10 units in population coverage with Family Health teams generates:
    1. for years not affected by the Covid-19 pandemic, an average decrease of 1.087 units in the fertility rate of women under 20 years of age in the municipality;
    2. for years affected by the pandemic, an average decrease of 0.833 units in this same rate.
- For a municipality with HDI-M in the 0.75 quantile of the HDI-Ms of Brazilian municipalities (from 0.718), an increase of 10 units in population coverage with Family Health teams generates:
    1. for years not affected by the Covid-19 pandemic, an average decrease of 0.692 units in the fertility rate of women under 20 years of age in the municipality;
    2. for years affected by the pandemic, an average decrease of 0.438 units in this same rate.

Furthermore, for the entire period of 2018-2021 - as it can be seen in **Table 28-**, an increase of 10 units in the PCP coverage in a municipality with HDI-M in quantile 0.25 generates a decrease of 1.451 units in the fertility rate of women under 20 years of age. For a municipality with a median HDI-M, this increase of 10 units in PCP coverage generates a decrease of 0.960 units in the fertility rate of women under 20 years of age. Finally, in a municipality with HDI-M in quantile 0.75, an increase of 10 units in the PCP coverage generates a decrease of 0.565 units in the fertility rate of women under 20 years of age in the municipality.

```r
## Creating a dataframe with the mean variations of the predicted fertility rates
## for each value of HDI-M when the primary care coverage is increased by 10
df_variacao_sem_pandemia <- df_newdata_completo |>
  group_by(idhm, pandemia) |>
  mutate(variacao = (predicoes - lag(predicoes))) |>
  ungroup() |>
  group_by(idhm) |>
  summarise(variacao_media = round(mean(variacao, na.rm = T), 3)) |>
  arrange(idhm)

kbl(
  df_variacao_sem_pandemia,
  align = "cc",
  col.names = c("HDI-M", "Mean variation on the predictions"),
  caption = "Average variation in predicted fertility rate values caused by an
    increase of 10 units in Primary Care coverage for each fixed HDI-M value."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 28: Average variation in predicted fertility rate values caused by an increase of 10 units in Primary Care coverage for each fixed HDI-M value.

HDI-M	**Mean variation on the predictions**
0.599	-1.451
0.665	-0.960
0.718	-0.565

References

R. A. Rigby, and D. M. Stasinopoulos. 2005. "Generalized Additive Models for Location, Scale and Shape,(with Discussion)." *Applied Statistics* 54.3: 507–54.

Stasinopoulos, Mikis D, Robert A Rigby, Gillian Z Heller, Vlasios Voudouris, and Fernanda De Bastiani. 2017. *Flexible Regression and Smoothing: Using GAMLSS in r*. CRC Press.